

MS ACCESS

2007

DIT Part 2nd



By: Sardar Azeem

PICT COMPUTER CENTER LINK ROAD
ABBOTTABAD(WWW.PICTCOMPUTERATD.COM)

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Course Overview

Discussion

The interface changes from Access 2003 to Access 2007 are significant. The familiar toolbars and menus are gone. The Office Button, Quick Access toolbar, Mini-toolbar, View tools and the Ribbon take their place. The prerequisite to this course, "Overview of Microsoft Office 2007," introduces you to these elements, which are common to most of the Office 2007 programs. These common elements will not be reviewed here.

The good news is that the interface changes are designed to make it easier to locate the commands that you need to do your work in Access. For experienced users, it will require a period of adjustment. We are in this together. Let us begin our adjustment now.

The New Database Format

Access 2007 will open, work with and save databases in Access versions 2000, 2002-2003 and 2007. However, if the database is not in the Access 2007 format, it cannot take advantage of the new features.

It is important to note that a database in the new Access 2007 file format (.accdb) cannot be opened in earlier versions of Access.

If the database is shared with other individuals, ensure that all of them are using Access 2007 before you convert the database to the Access 2007 format. See page 104 of this manual for conversion instructions.

Note that this new format does not support user-level security. If user-level security is required, use a file format from an earlier version. The new format also does not support replication.

The new Access 2007 format supports new features, such as:

- tabbed views
- new views of objects (split-form view, Layout view, Report view)
- multivalued lookup fields
- attachment data type
- Total rows in Datasheet view of a Table, Form or Query
- more compact database file

Course Topics

- The New Interface
 - The Getting Started window
 - Open an existing database
 - The Ribbon
 - Open a database object
 - Customize for overlapping windows or tabbed views
 - Search for an object
 - Navigate in an open table or form
 - Switch views
 - Total row in Datasheet view
 - Close objects
 - Close a database
- Databases
 - Create a new database
 - Database format
- Tables
 - Create a table
 - Import data
 - Multivalued lookup fields
 - Filtering (AutoFilter)
 - Copy, save and delete objects
 - Printing
 - Table relationships
 - Validation rules
- Forms
 - Create a basic Form
 - Edit a Form
 - Use the Form wizard
 - Filter
 - Printing
- Queries
 - Create a query
 - Object dependencies
 - Export data to Excel and Word
 - Create a PDF file
- Reports
 - Use the Report wizard
 - Edit a Report
 - E-mail a Report
 - Use Report View
- Security
 - Security

Conventions Used in This Manual

In each lesson, there can be:

- A list of topics
- A general discussion of the lesson topics
- A review of several topics
- One or more exercises
- One or more tips
- One or more notes



A triangular bullet is used for the **list of lesson topics**.



A topic heading is denoted by the Steps graphic, if a list of general **Steps** is given. Back at your desk, you may refer to these Steps as a basic outline of what to do.



An **Exercise** heading is denoted by the Runner graphic. These classroom exercises provide practice and more detailed information about the topic(s) being covered.

Tip – Open a File
The shortcut
CTRL + O
will display
the Open
dialog box.

A **Tip** box is available to offer a helpful hint or a caution about the topic that is being reviewed.

Note:

A **Note** provides supplementary information about the current topic

Lesson 1: The New Interface

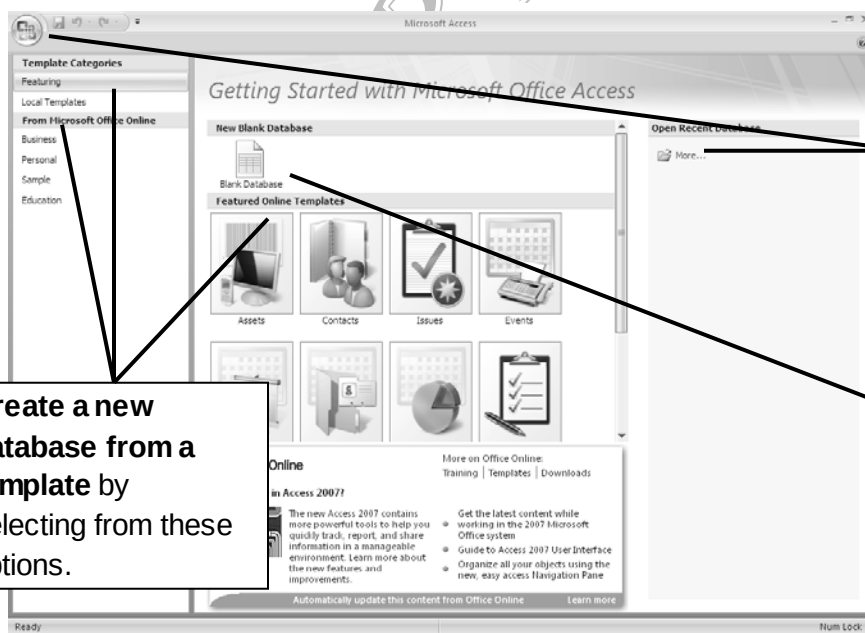
Lesson Topics

In this lesson, you review the appearance and functionality of the new windows and navigate in an existing database. You will:

- ▶ Start Access and review the Getting Started window
- ▶ Open an existing database
- ▶ Review the Ribbon tabs
- ▶ Open a database object
- ▶ Customize Access for overlapping windows or tabbed views
- ▶ Search for an object
- ▶ Navigate in an open table or form
- ▶ Switch views for the object
- ▶ Work with a Total row in Datasheet view
- ▶ Close the object
- ▶ Close the database

The Getting Started Window

When Access is opened, the Getting Started window is displayed. This initial window is notably different from that of Access 2003.



Open an Existing Database

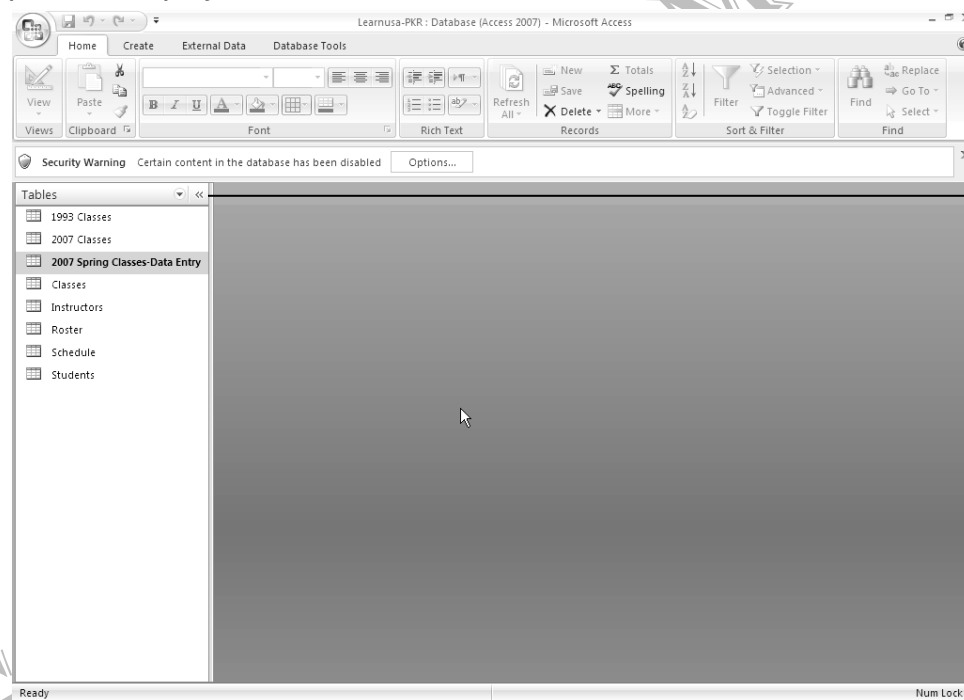


1. Start Access 2007.
2. In the Open Recent Database pane on the right side of the Getting Started window, click once on the  tool or on one of the filenames in the Open Recent Databases list. The Open dialog box is displayed.

Alternatively, click on the Office Button in the Getting Started window. Select Open **OR** select one of the filenames in the Recent Documents list.

3. Navigate to the folder that contains the database and double click on its filename. In class, open **Learnusa.accdB** in the Access folder.

The Access application window changes its display and the database Navigation pane is displayed.



The Navigation pane replaces the Database window.

Note: The extension **.accdb** is used for Access 2007 databases. As we transition from Access 2000 or 2003 databases to Access 2007, note the title bar for the window. If the database is in an earlier format, it is stated in the title bar.

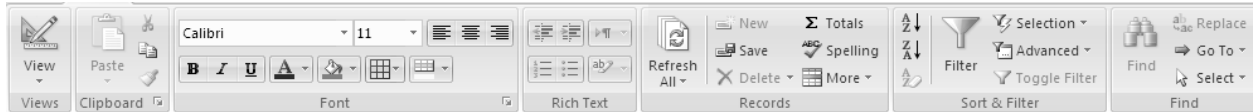
Learnusa4 : Database (Access 2000 file format) - Microsoft Access

Access will save the database in the same format, unless you specifically select to save in another format using the Save As command on the Office Button.

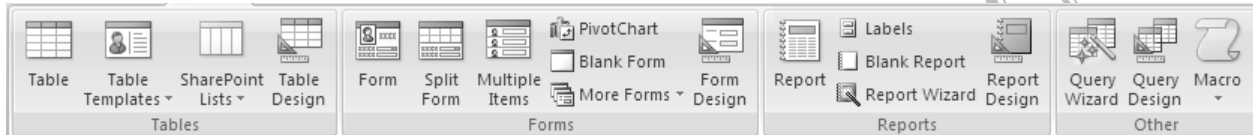
The Ribbon

The Ribbon in Access has four initial tabs: Home, Create, External Data and Database Tools.

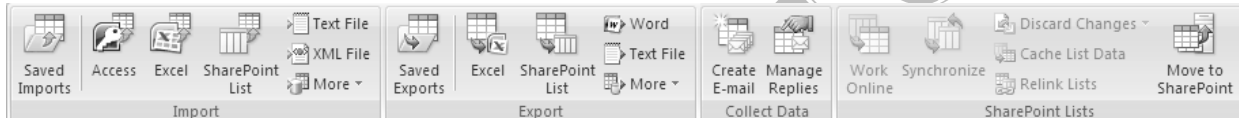
Home tab



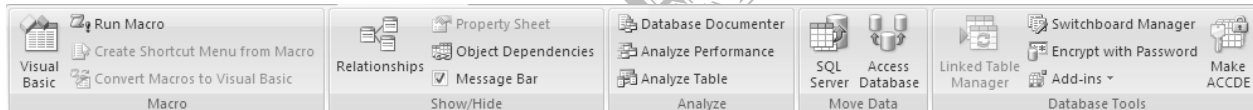
Create tab



External Data tab



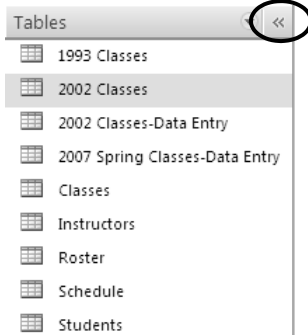
Database Tools tab



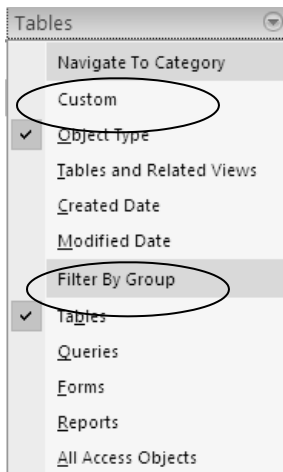
Open a Database Object



1. In the Database window, click on the down arrow in the title bar.



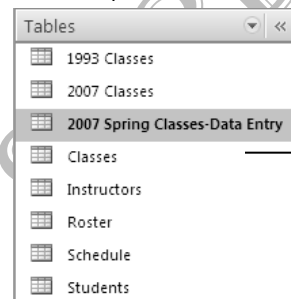
A list of database objects, properties and related choices is displayed.



Select a **category** for the list.

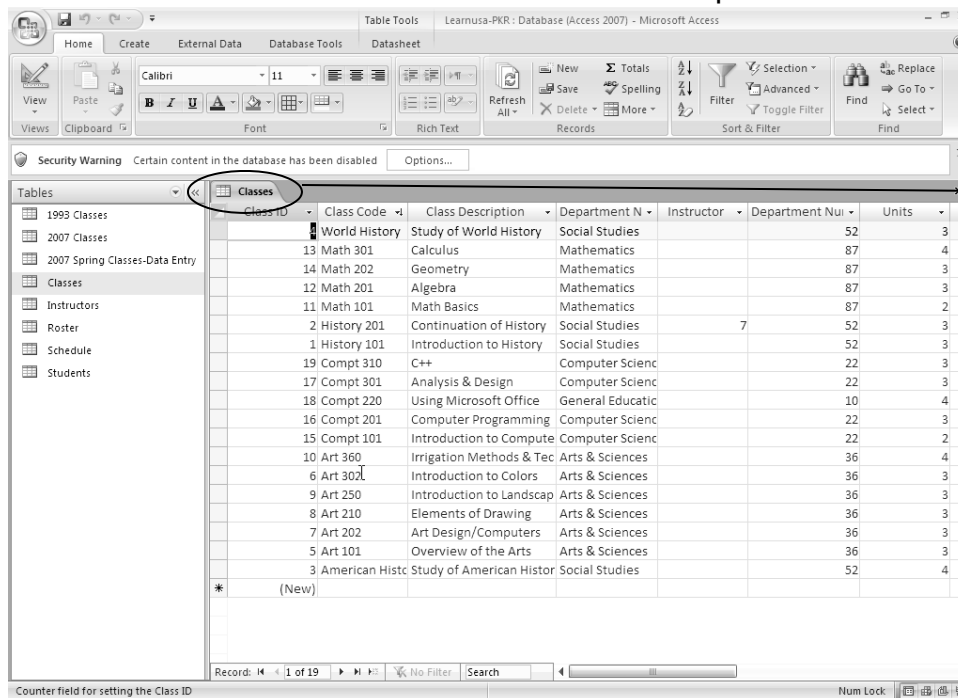
Then, select the **group** of items that you want to see in the list.

2. Select the Category.
In class, select **Object Type**.
3. Click on the down arrow and select the type of objects to appear in the list.
In class, select **Tables**. A list of the objects is displayed in the Navigation pane.



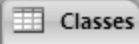
The list includes the objects for the category and object type that you selected in steps 1 and 2.

4. Double click on the icon for the **Classes** table to open it.



When an object is opened in an Access 2007 database, it has an identifying tab.

5. Note that the object opens to the right of the Navigation pane. This leaves the Navigation pane visible while the object is open.

6. Note that the object has a tab  to identify it.

7. Click on the down arrow in the title bar of the Navigation pane.

Tables

8. Select another object from the list.

In class, select **Forms**.

9. From the list of forms, select **Classes**. The Classes form is displayed.

10. Note that the tabs provide easy navigation between the two open objects. Click on a

tab to view that object.



11. The Navigation Pane can be hidden.

Click on the Open/Close tool  in its title bar to hide and to unhide the Navigation Pane.

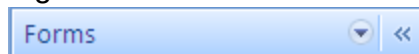
OR

Press the F11 to hide and then to unhide the Navigation Pane.

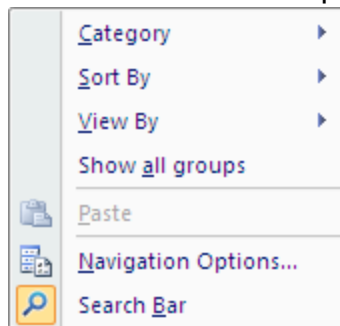
An icon indicates the type of object:

-  Table
-  Select Query
-  Form
-  Report

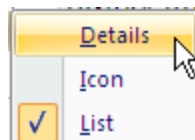
12. Right click on the title bar of the Navigation pane.



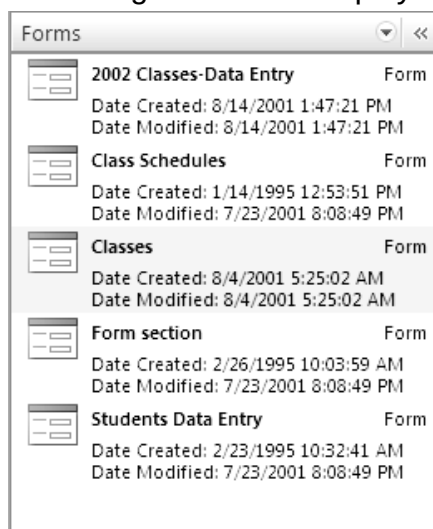
A shortcut menu is displayed.



13. Point to View By and select Details from the list.



14. The Navigation Pane display shows the details for each object.



Customize for Overlapping Windows or Tabbed Documents



When objects are opened, the objects' windows can be displayed:

- On top of one another. This overlapping of windows was the method used in Access 2003.

OR

- As windows with tabs that are placed in the order in which you opened the object. This view is available only for Access 2007 databases.

In step 4 of the previous section of this lesson, you saw the default for Access 2007 databases - tabbed windows. If you prefer the overlapping windows, follow these steps to change **this database only**.

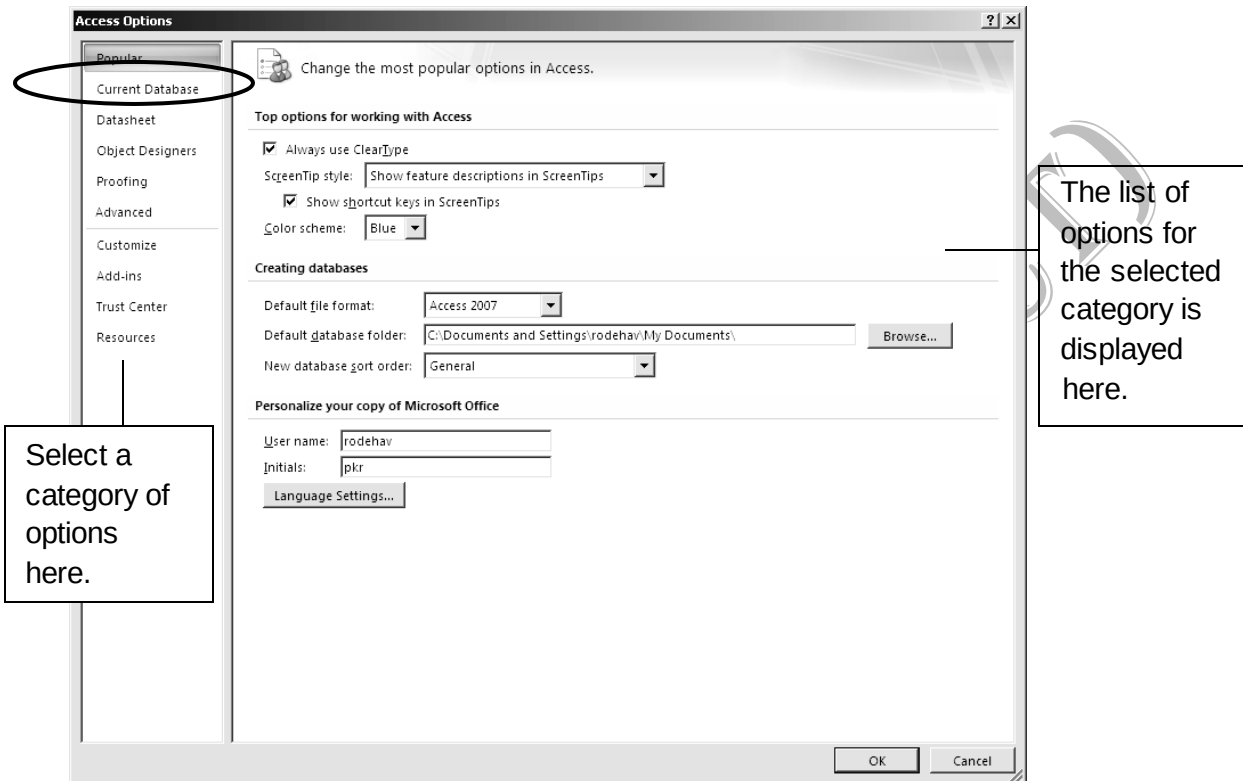
1. Open the database in which you want the overlapping windows display.
2. Click on the Office Button . The following list of commands is displayed.



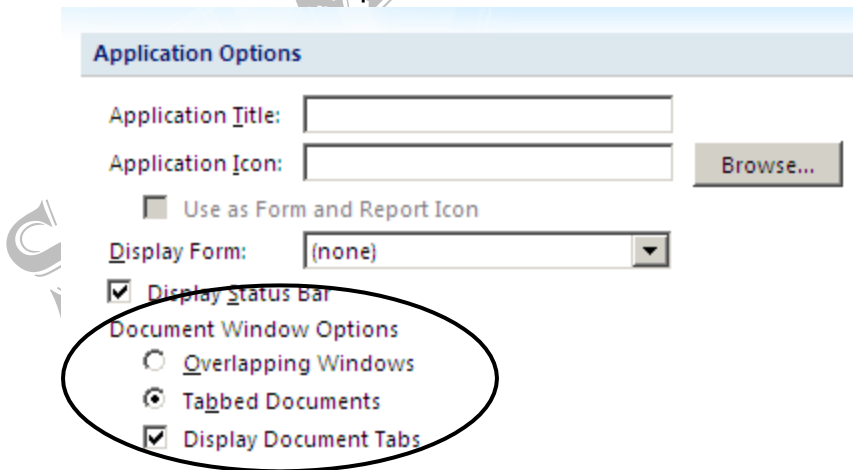
Tip – Overlapping Windows

By default, Access 2007 databases open with tabbed view and Access 2000, 2003 databases open with overlapping windows. You can change the default using the Access Options button.

3. Click on , which is in the lower right corner of the command list. The dialog box is displayed.

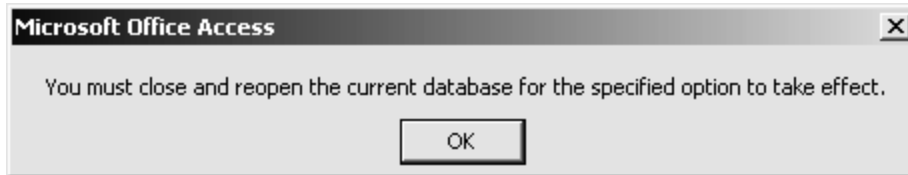


4. Select **Current Database** in the list of categories. The list of options in the right side of the dialog reflects your category choice.
5. In the **Application Options** on the right side of the dialog, locate the Document Window Options area.



6. Select Overlapping Windows (or Tabbed Documents, if you changed this previously, and want to change it back. If you select Tabbed Documents

7. Click on OK. A message is displayed.



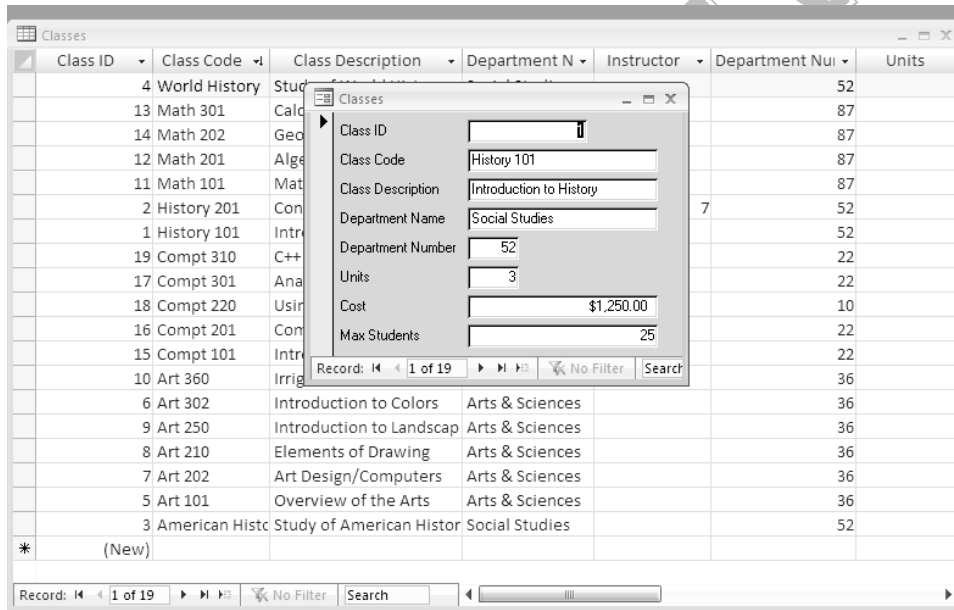
8. Click on OK, again.
9. If you want the new choice to take affect now, close and reopen the database.

To close and reopen the database:

Click on the Office Button and select Close Database.

In the Getting Started window of Access, select the database from the Open Recent Database list.

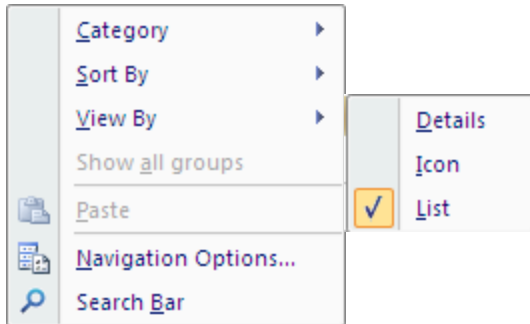
This is an example of overlapping windows:



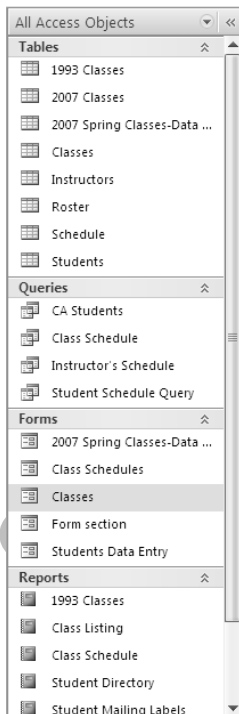
Exercise 1 - The Navigation Pane Display



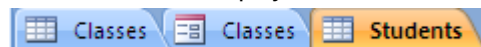
1. Right click on the title bar of the Navigation Pane. A shortcut menu is displayed.
2. Select View and then List.



3. Click on the down arrow in the title bar of the Navigation Pane
4. In the Filter by Group area of the list, select **All Access Objects**. All the objects in the database are listed and categorized by type.



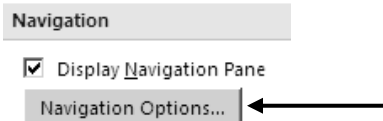
5. Double click on the icon for the **Students** table to open it. Three objects are open.
6. Follow the instructions beginning on page 14 to switch back to Tabbed Documents.
7. Open the Classes table, Classes form, and Students table. The tabs displayed are



Enable and Use the Search Box in the Navigation Pane



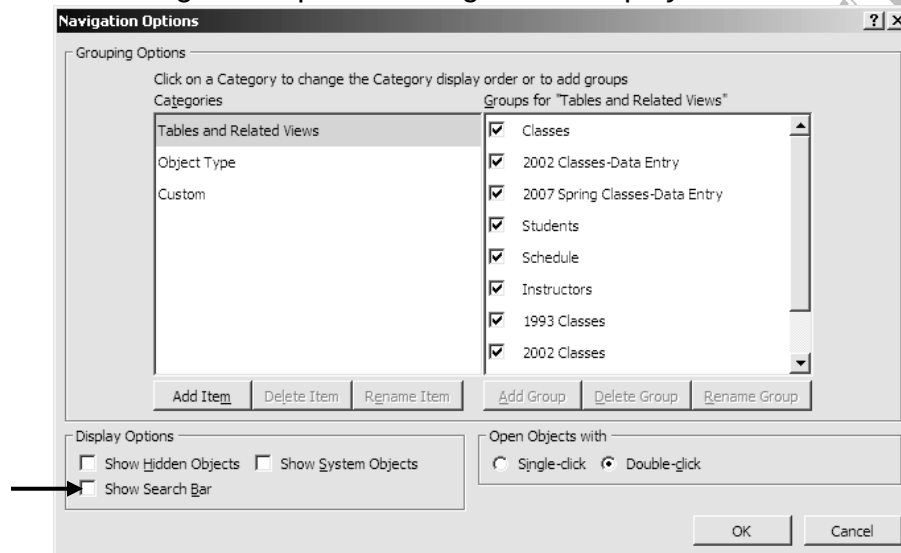
1. With the database open, click on the **Office Button**.
2. Select **Access Options**. The dialog box is displayed.
3. Select **Current Database** in the left column.
4. In the Navigation area of the right column, click on the Navigation Options button. .



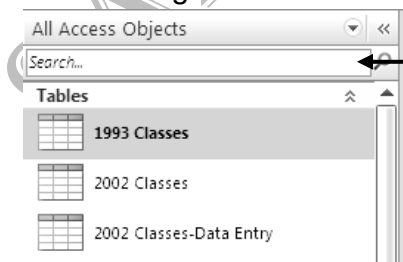
Tip – Navigation Options

The Navigation Options dialog box also is available by right clicking on the Navigation Pane's title bar.

The Navigation Options dialog box is displayed.



5. Mark the checkbox for **Show Search Bar** in the Display Options area.
6. Click on OK.
7. Click on OK, again. The Search box is added below the title bar of the Navigation Pane.



The Search box uses the logical operator **Contains**. It locates any object where the name contains the characters that you typed.

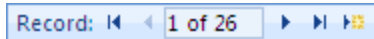
8. Try it out. Start to type Class in the Search box.
9. Backspace to erase the typing and type Sch.
10. Backspace to clear the Search box and display all objects.

Navigate in an open table or form



To change the position of the cursor in a table, use the same methods as in previous versions of Access.

- Press the Tab key to move forward a field at a time.
- Hold down the Shift key while you press Tab to move backward a field at a time.
- Click in a field to place your cursor in that field.
- Use the navigation buttons in the lower left corner of the window.



With the tabbed objects, click on a tab to place that object on top.

Switch views

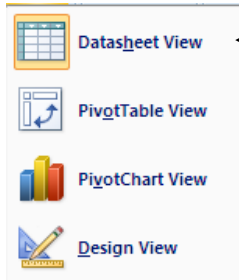


Each object has at least two views. In one view, the data is displayed. In the second view, the object's design is displayed. Additional views are available for most objects.

1. Click on the **Home** tab if it is not the active tab of the Ribbon.
2. The **Students table** should be the active object.



3. Click on the View tool. A list of views available for that object is displayed. For a table, the list shown is:



Tip – Switch Views The list of views also is available by right clicking on the object's name.

Note that the icon for the current view is highlighted in the list.

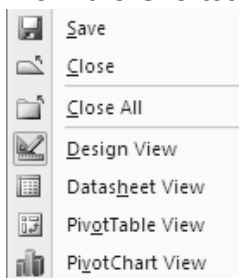
4. Click on the view that you want.

In class, click on  **Design View**. The Design view of the table is displayed.

Field Name	Data Type
StudentID	Text
FirstName	Text
MiddleName	Text
LastName	Text
Address	Text
City	Text
State	Text
PostalCode	Text
PhoneNumber	Text
DOB	Date/Time
Sex	Text
Major	Text

Field Properties	
General Lookup	
Field Size	12
Format	
Input Mask	000\-\00\-\0000\;
Caption	Student ID
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	No
Indexed	Yes (No Duplicates)
Unicode Compression	Yes
IME Mode	No Control
IME Sentence Mode	None
Smart Tags	

5. Another method to switch views of an object is to right-click on the object's tab. From the shortcut menu that is displayed, select the view you want.



In class, return to the  **Datasheet View**.

Classes		Students	
First Na	Middle Na	Last Nai	Address
Justin	Mark	Mester	3 Anchor Avenue
Lauri	Marie	Jones	188 Peanut Place
Norma	Kay	Bestic	10382 Pine Avenue

Total Row in Datasheet View

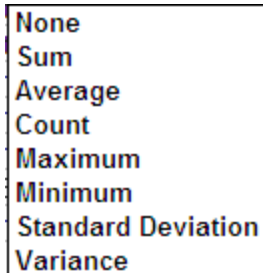
Discussion

You may be familiar with the Sum tool  in Excel. In Access, it is named the Totals tool. This tool is available now on the Home tab in Access, when the datasheet view of an object is displayed. The Totals tool displays a total row below the entry row of the datasheet. Click in any field in the Total row and a down arrow appears.

The list on the down arrow displays functions that are appropriate for the data type in that field.

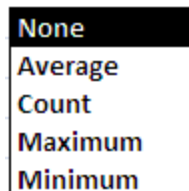
In text fields, only the Count function is available.

In numeric fields, the list contains



- None
- Sum
- Average
- Count
- Maximum
- Minimum
- Standard Deviation
- Variance

In date fields, the list contains



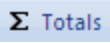
- None
- Average
- Count
- Maximum
- Minimum

Steps



1. With the Datasheet view of an object displayed, click on the Home tab of the ribbon.

In class, use the Students table.

2. Click on the  **Totals** tool. A total row is displayed at the bottom of the display.

This row actually is below the data entry row of the datasheet. However, it displays on the screen no matter which records are in view.

Student ID	First Na	Middle Na	Last Na	Address	City	Stz	Postal Cc	Phone Num
111-32-4832	Justin	Mark	Mester	3 Anchor Avenue	Huntington Be	CA	92545-4651	(714) 678-9522
134-56-7894	Lauri	Marie	Jones	188 Peanut Place	Los Alamitos	CA	90807-	(310) 648-5556
134-56-7898	Norma	Kay	Bestic	10382 Pine Avenue	Long Beach	CA	90809-4624	(310) 579-4987
164-89-4562	Diane	Patricia	Sunter	123 Frost Avenue	Houston	TX	53423-1854	(214) 677-9752
164-89-7562	Tracy	L.	Anderson	1772 Dester Avenue	Portland	OR	97208-4029	(503) 794-5378
246-84-8975	Art	L.	Christen	18277 Rose Lane	Phoenix	AZ	85642-6789	(602) 497-8956
334-56-7895	Mariana	Marie	Gonsolaz	555 Trest Street	Huntington Be	CA	92545-4988	(714) 860-6452
444-54-5789	Jerri	T.	Lusten	2283 Orange Ave.	Alta Loma	CA	91526-4894	(909) 456-1567
444-56-4897	Joseph	J.	Jester	166 8th Place	New York	NY	85642-6788	(602) 887-5975
446-48-6156	Melissa	Kathryn	Maloney	321 Santa Street	Santa Ana	CA	91256-4511	(714) 648-7985
446-57-8941	Alyson	Lynne	Becktor	12 Aviation Blvd.	El Segundo	CA	90845-4648	(310) 468-7954
464-59-7561	Michael	Raymond	Uher	65 Ski Place	Signal Hill	CA	90806-1875	(310) 425-9756
467-85-2678	Ruth	Louise	Rotterson	34 Lester Street	Phoenix	AZ	85642-6789	(602) 784-8975
467-89-4152	Leslie	Marie	Chin	182 1st Street	Los Alamitos	CA	90712-3468	(310) 678-9574
497-55-1967	Patricia	M.	Crutz	14 Mabaker Drive	Erie	PA	16501-9631	(814) 114-5784
549-72-8265	Justin	Lawrence	Johnson	773 5th Street	Cypress	CA	90705-1645	(310) 975-6425
555-64-8675	Samuel	Dean	Thompson	19 Apple Drive	Riverside	CA	95446-4578	(909) 645-6785
558-97-5648	Amy	Laurie	Justines	14 Trade Avenue	New York	NY	10199-	(212) 648-9755
564-89-7526	Jerry	A.	Lester	1039 8th Street	Long Beach	CA	90806-	(310) 564-8975
564-89-7564	James	L.	Jemmison	72 Lost Terrace Rd	Redondo Beach	CA	90456-7489	(310) 664-5675
567-94-5327	Michael	Kenneth	Marlow	1029 Lanser Avenue	Columbia	SC	29292-4655	(803) 678-9204
645-84-8975	Mary	Ann	Koite	102 Peanut Place	Los Alamitos	CA	90807-1645	(310) 648-6755
Total								

3. Scroll to the right to view the DOB field.
4. Click in the Total row for that field. A down arrow is displayed.
5. From the list at the down arrow, select **Minimum**.
The earliest birth date is displayed. **12-May-75**
6. Click in the Total row for the Phone Number field and select **Count**.
The number of values in this field is displayed. **26**

Close Objects

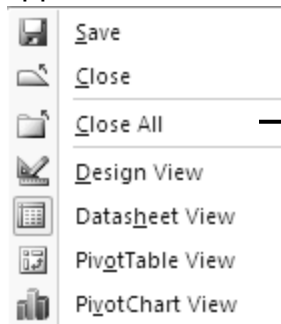


To close a database object, do one of the following:

- Click on the  in the upper, right corner of the object's window.



- Right click on the object's tab and select  **Close** from the shortcut menu that appears.



Tip – Close All Open Objects
Use the Close All command to close all the open objects in the database. This command does not close the database.

- Press CTRL + F4 on the keyboard.

Close a Database



- Click on the **Office Button** .
- Select **Close Database** from the list of commands. The database closes and Access remains open. The Getting Started window of Access is displayed.

Lesson 2: Databases

Lesson Topics

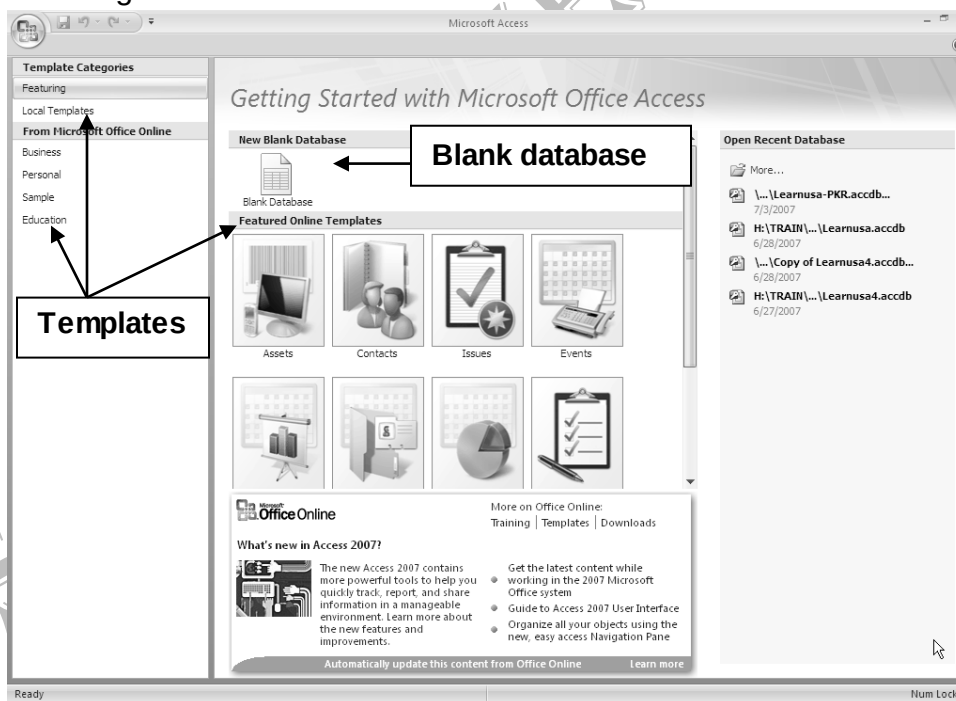
In this lesson, you review the format of the database file and how to create a new database. You will:

- ▶ Create a blank database
- ▶ Review database formats
- ▶ Review saving the database in an earlier Access format
- ▶ Change the default file format

Create a new database

A new database is created from the Getting Started window in Access 2007 by:

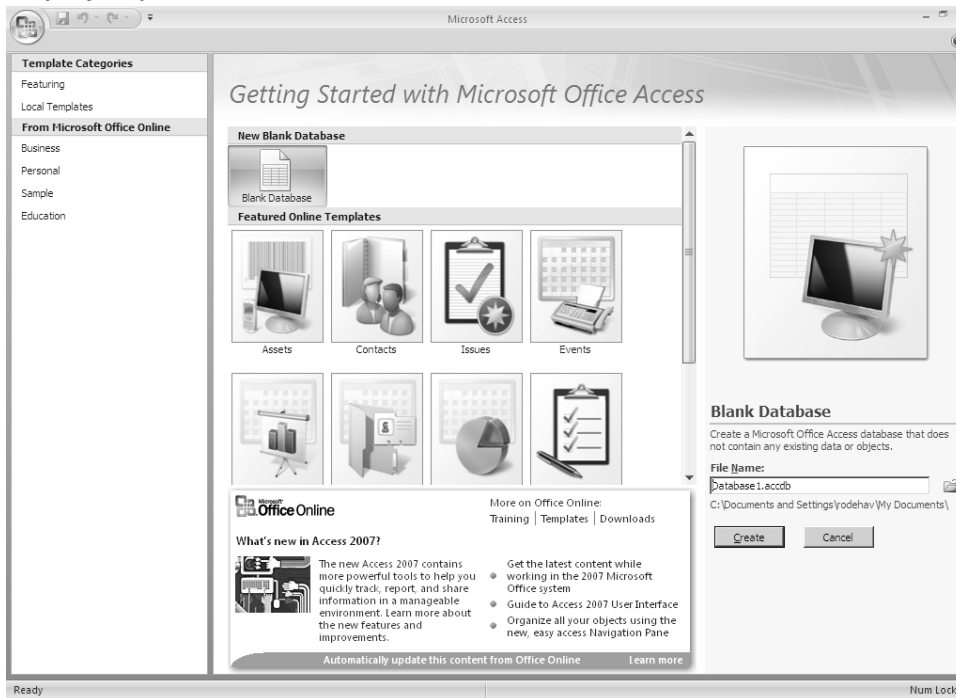
- Using a **template**
OR
- Selecting to **create a blank database**.



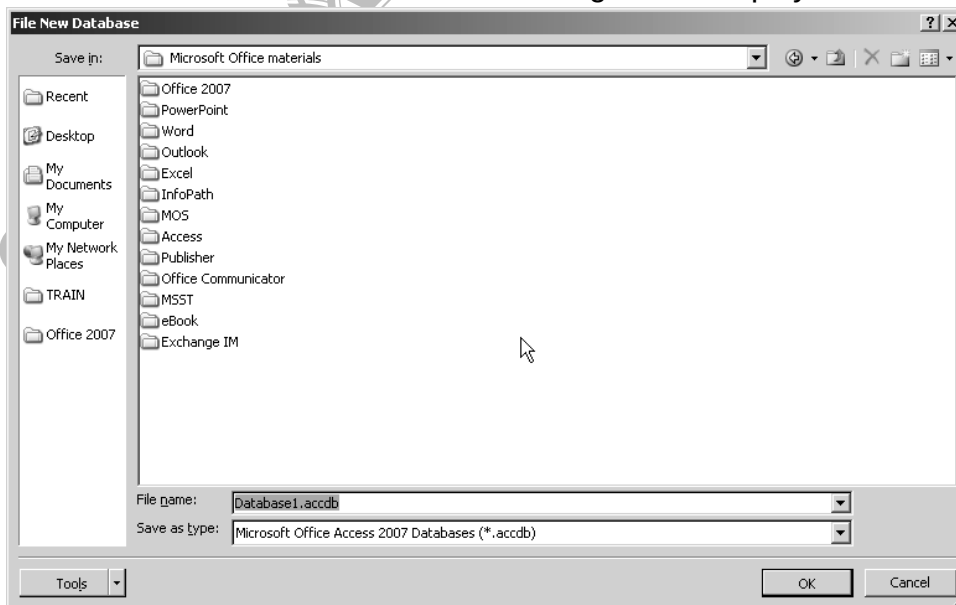
Create a Blank Database



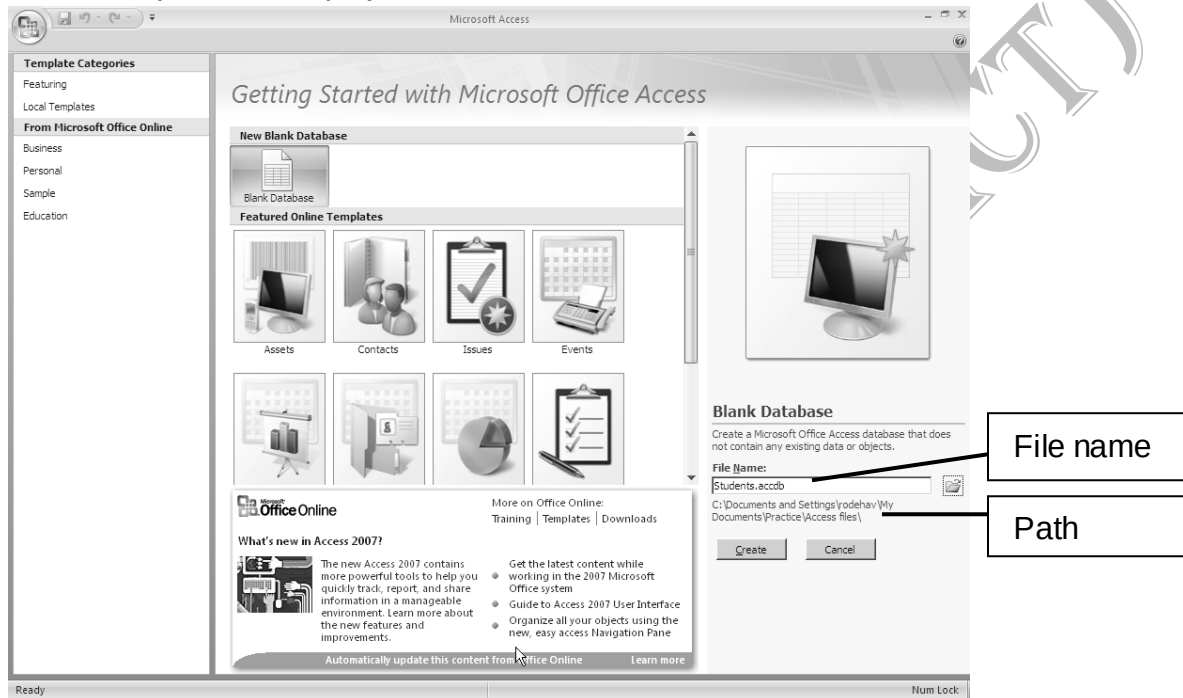
1. Click on **Blank Database**. The right side of the Getting Started window is changed to display a pane in which to name the new database.



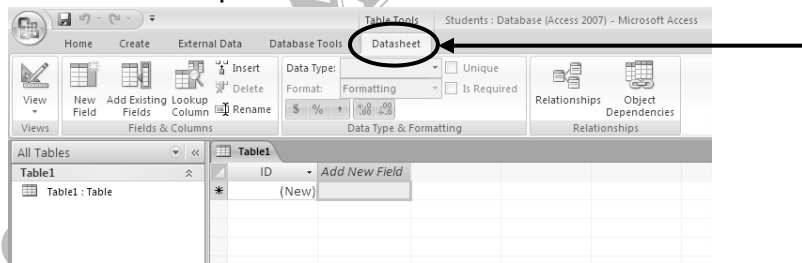
2. Click on the Browse tool , which is indicated by the arrow in the above screenshot. The File New Database dialog box is displayed.



- At Save in, navigate to the drive and folder in which to store the new database. In class, follow the instructor's directions.
- In the File name edit box, type a name for the database. In class, type **Students**.
- Click on OK. The Getting Started window of Access is displayed, again. The file name and path are displayed.



- Click on **Create**. The Ribbon and the Navigation pane are displayed. A new blank table is open in the datasheet view.



- Note that the Datasheet contextual tab is displayed in the Ribbon.

Database formats

Discussion

Access 2007 recognizes and will open databases in the formats of earlier versions of Access since 2000. The format will display in the title bar of the application.

Students : Database (Access 2007)

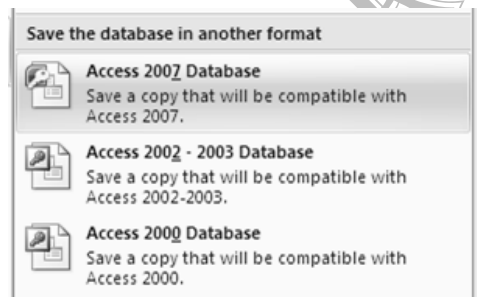
Extension	Access Version
.accdb	Access 2007
.mdb	Access 2002-2003
.mdb	Access 2000

An Access 97 database can be opened, but it must be converted to one of the above formats in order for Access to work with it. No design changes can be made in the database until it is converted to the Access 2000 format or later. When you try to open an Access 97 database, a dialog box is displayed to help with the conversion.

By default, when a new database is created in Access 2007, it uses the new format with the .accdb file extension.

Save a database in an earlier Access format

1. Click on the Office Button.
2. Point to Save As. When the Save As options are displayed, select from options at Save the database in another format.

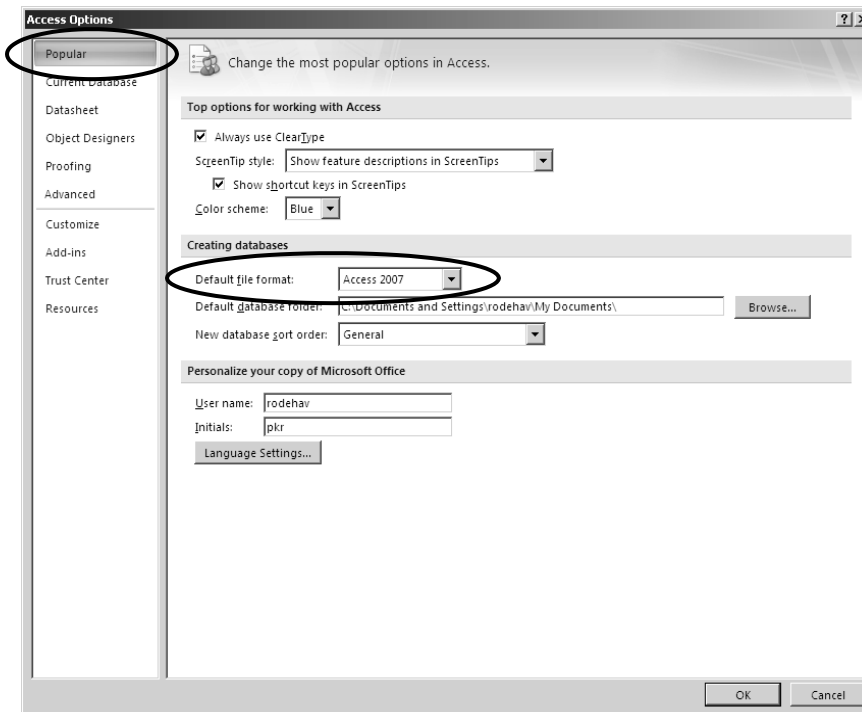


Change the default file format

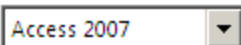


If all the databases that you create should be in an earlier format of Access, change the Default File Option with these steps.

1. Click on the **Office Button**.
2. Select  **Access Options** from the lower right corner of the command list. The Access Options dialog box is displayed.



3. Select the **Popular** category on the left pane of the dialog.
4. In the right pane of the dialog, click on the down arrow at

Default file format: 

5. Select from the list of available file formats.

Access 2000
Access 2002 - 2003
Access 2007

In class, leave the default of Access 2007.

6. Click on OK to save the changes.

Lesson 3: Work with Tables

Tables are the only database objects that store data. All other objects, such as forms, queries and reports are ways to review the data in a meaningful way.

Lesson Topics

In this lesson, you create tables, work with the data and work with the database objects. You will:

- ▶ Create a table
- ▶ Import Excel data
- ▶ Create Multivalued Lookup fields in a table
- ▶ Filter data in a table
- ▶ Copy, save and delete database objects
- ▶ Print an open object
- ▶ Print table and other object definitions
- ▶ Establish table relationships
- ▶ Review validation rules

Create a Table

1. Click on the **Create** tab.



2. Click on . A new table is created and displayed in Design view. This view is the same as in Access 2003.

The screenshot shows the 'Table1' window in Access. It has a table grid with three columns: 'Field Name', 'Data Type', and 'Description'. Below the grid is the 'Field Properties' section, which is divided into 'General' and 'Lookup' tabs. The 'General' tab is currently selected and is empty. The 'Lookup' tab contains a text box with the message: 'A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.'

3. Add the Field Names and Data Types in the top pane.
In class, add these Field Names and Data Types.

Field Name	Data Type
EmplID	Text
LName	Text
FName	Text
CampusPhone	Text
Email	Hyperlink

4. Click in a Field to display its Field Properties in the bottom pane.
5. Edit the Field Properties as appropriate.
In class, set the following field properties.

EmplID	Size – 8	
LName	Size – 40	Required – Yes
FName	Size – 30	Required – Yes
Campus Phone	Size - 14	



6. Select a primary key field by clicking in the field and selecting  .
In class, set the **EmplID** field as a primary key.
7. Click on the Save tool  in the Quick Access toolbar to save the design changes and give the table a name.
In class, name the table **ContactStudent**.

Tip – Naming Objects
It is best not to include spaces in the names of objects in the database.

8. In class, modify the table by editing the field properties of the **CampusPhone** field.
- Click in the CampusPhone field.
 - In the bottom pane, click in the Input Mask field property.
 - Use the expression builder tool  in the Input Mask property to start the wizard that will assist you. The wizard is the same as that in Access 2003.
9. Click on the **Save** tool  to save the design changes.

Note: There is a new data type in Access 2007. It is the **Attachment data type**. This allows you to store entire files in your Access database as attachments to a field in a table. A single field of the Attachment data type can store multiple attached files.

This data type should be used with caution. Attaching data files to the database will increase its size and may burden the database.

Import Data

Discussion

An import copies data, which is stored externally, into an Access database. Data may be imported from:

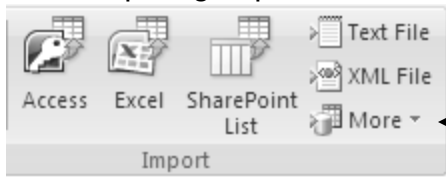
- Another Access database
- Excel or Lotus 1-2-3
- A delimited text file
- A fixed-width text file
- An XML document
- An HTML document
- A dBASE or Paradox file

Tip – Export Data For information about exporting data, refer to page 72.

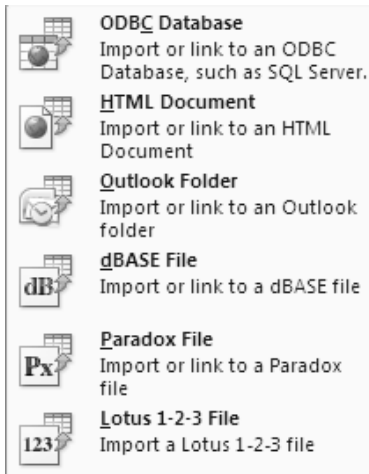
Import data



1. Open the database that will receive the imported data.
2. Select the External Data tab.
3. In the Import group, click on the tool for the type of data to be imported.



Click on the More down arrow to select from the following types of imported data.



4. Follow the instructions in the wizard.

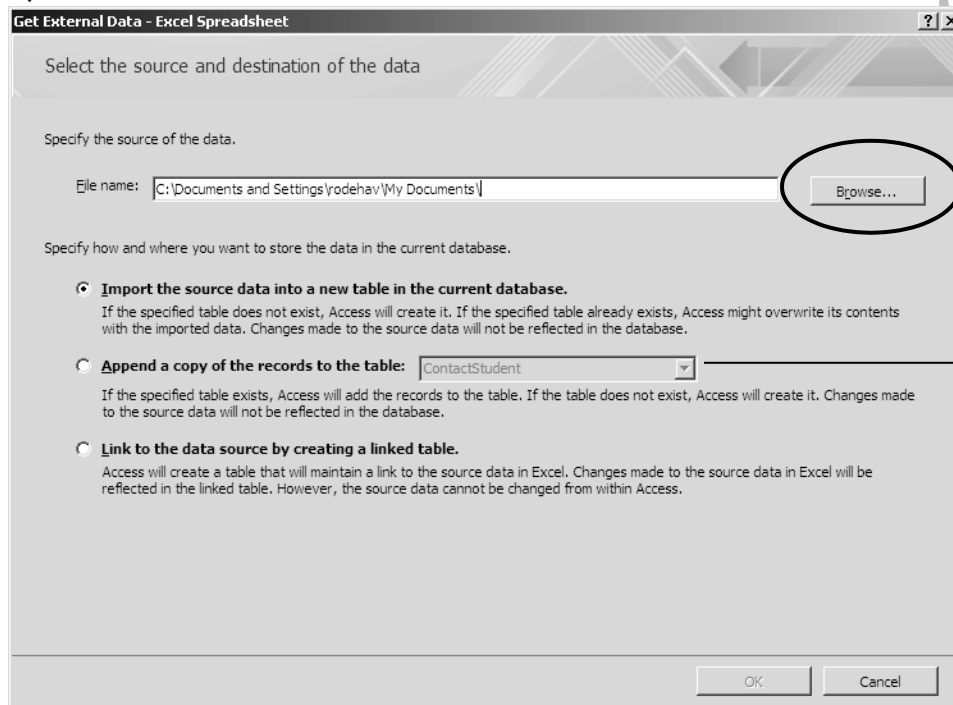
Exercise 2 - Import Excel data



1. With the Students database open, click on the **External Data** tab.



2. In the Import group, click on the **Excel** tool. The Get External Data – Excel Spreadsheet Wizard starts.

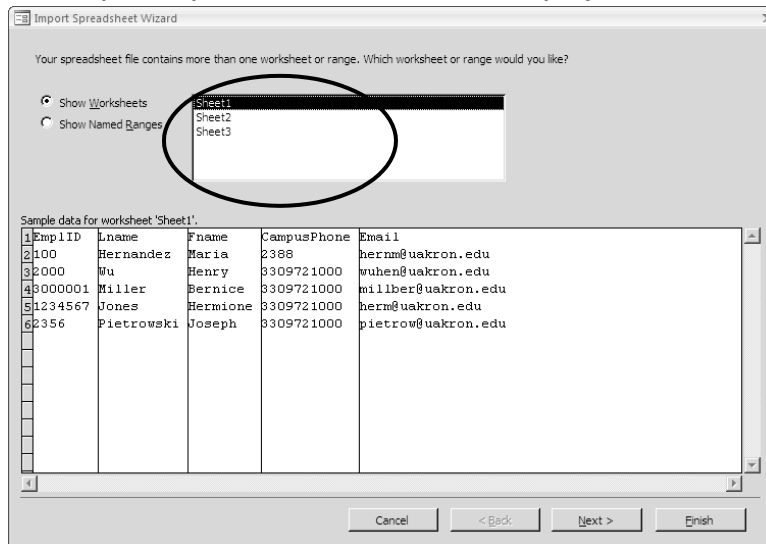


To append the data, the field names in the data file must match the field names in the table.

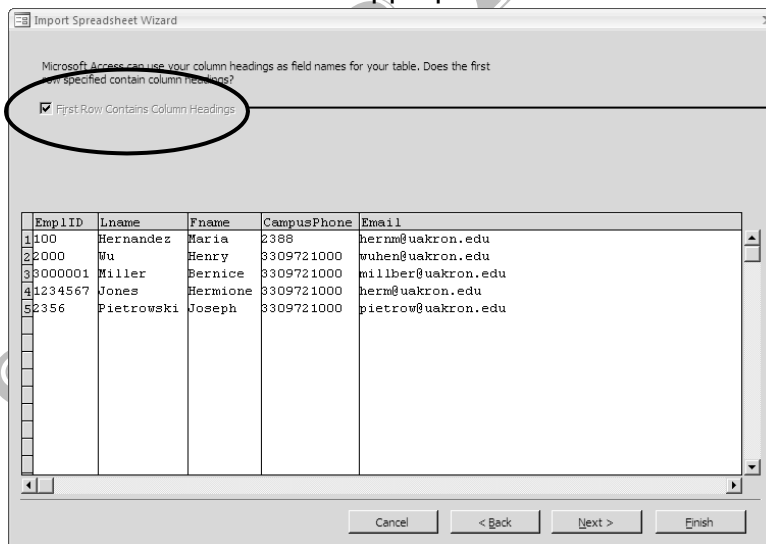
3. At "Specify the source of the data," click on **Browse...**. The File Open dialog box is displayed.
4. At Look in, navigate to the location of the Excel file that contains the data to be imported.
5. Click once on the name of the Excel file. In class, select **Import.xlsx**.
6. Click on **Open**.
7. At "Specify how and where you want to store the data in the current database," select the appropriate option button.
In class, select **Append a copy of the records to the table**. The down arrow for the list box will activate.
Be sure that ContactStudent is selected.
8. Click on OK.

9. If prompted that the table is open and needs to be closed and/or saved, select the responses to close and save the table.

The Import Spreadsheet Wizard is displayed.

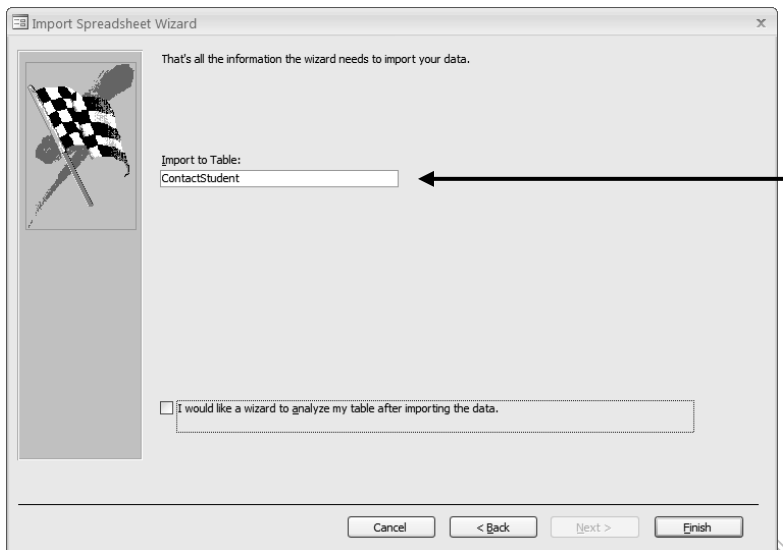


10. Select the sheet that contains the data. The data is displayed in the lower half of the dialog box. In class, select **Sheet1**.
11. Click on Next.
12. In this step, be sure that the checkbox "First row contains column headings" is marked or unmarked as appropriate.

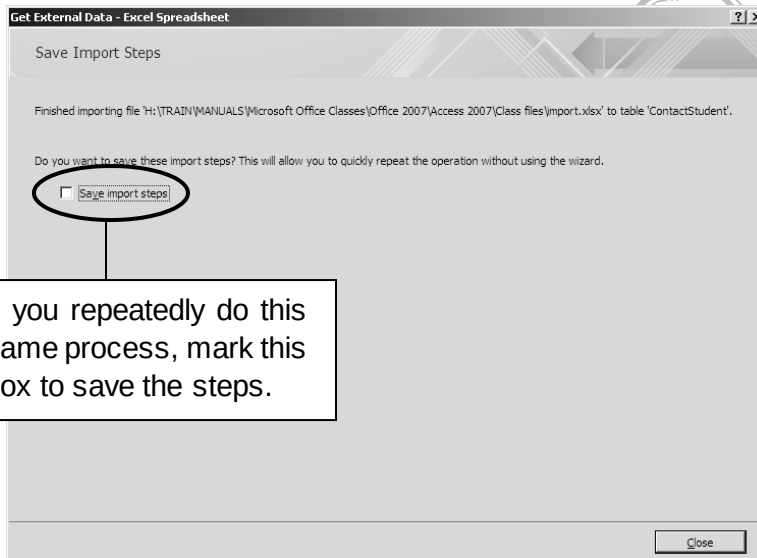


If you are appending the data, the checkbox will be marked and not available for editing.

13. Click on Next. The correct table name should be in the edit box.



14. Click on Finish. If the import was successful, the following dialog box is displayed.

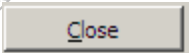


If you repeatedly do this same process, mark this box to save the steps.

Tip - Import data

It is easier to import data to a new table than it is to import data to append to an existing table. If the data is to be appended, review the original data to assure that the field names and data types will match that of the existing table.

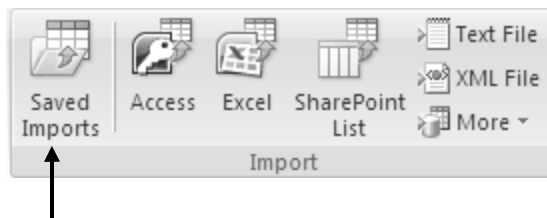
If there was a problem with the import, an error message is displayed. Review the message and take the appropriate action.

15. Click on .

16. Open the ContactStudent table to review the imported data.

ContactStudent				
EmplID	LName	FName	CampusPhoi	Email
100	Hernandez	Maria	() -2388	hernm@uakron.edu
1234567	Jones	Hermione	(330) 972-1000	herm@uakron.edu
2000	Wu	Henry	(330) 972-1000	wuhen@uakron.edu
2356	Pietrowski	Joseph	(330) 972-1000	pietrow@uakron.edu
3000001	Miller	Bernice	(330) 972-1000	millber@uakron.edu
*				

Note: If the box was marked to **Save import steps**, the saved steps are accessed on the External Data tab in the Import group. Click on the Saved Imports tool.



Multivalued Lookup Fields

Discussion

In Access, a Lookup Field is a column where the values are returned from another table or value list in the database. The person doing the data entry sees a list of values at a down arrow and selects a value from the list.

In Access 2003, only one value could be selected from the list. In Access 2007, databases in the 2007 format can have a Lookup Field set to accept multiple values.

Create a Multivalued Lookup Field



1. Create a table that will contain the Lookup values. Follow the instructions given earlier in this lesson.

In class, create the **AcademicPlans** table shown here.

ID	Text	Field size 10
PlanName	Text	Field size 70

Make the ID field the primary key.

2. Switch to the Datasheet view and add the data.

AcademicPlans	
ID	PlanName
100100	Dog Walking
200100	Chilling
300100	Beach Walking
400100	Chocolate Making
500100	Basket Weaving

Tip – Saving Records

A record is saved when you move off it, press Shift+Enter or click on the Save tool  in the Records group of the Home tab. Records also are saved if you close a table or form.

3. Close the table by clicking on its X.



4. Open the table in which to create the Lookup field.

In class, open the **ContactStudent** table, if necessary.

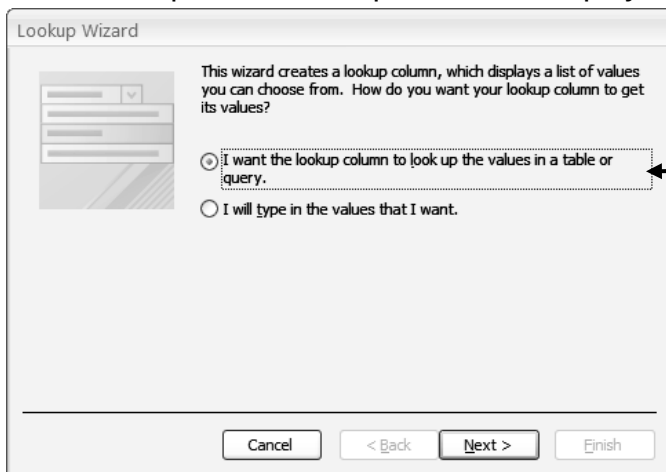
5. Switch to the Design view.

6. Add a new field by entering a field name.

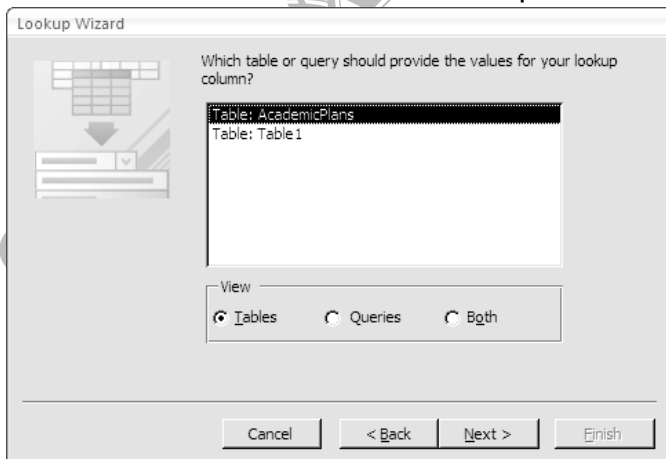
In class, type **AcademicPlan** as the fieldname

	Field Name	Data Type
	EmplID	Text
	LName	Text
	FName	Text
	CampusPhone	Text
	Email	Hyperlink
	AcademicPlan	lookup Wizard...

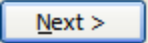
7. Select **Lookup Wizard** under Data Type.
8. Press the ENTER key.
9. The first step of the Lookup Wizard is displayed.

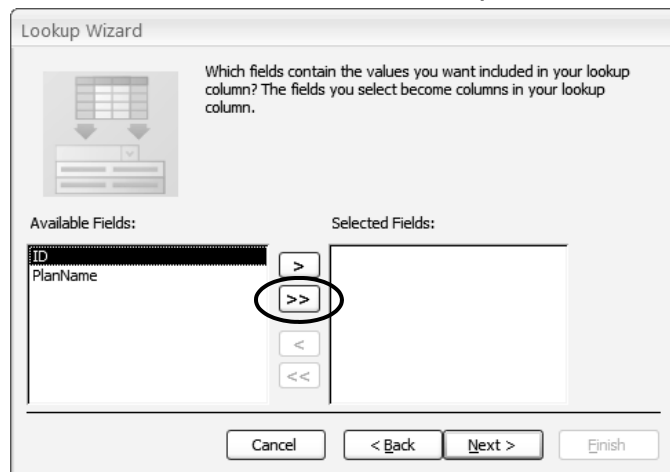


10. Select "I want the lookup column to look up the values in a table or query."
11. Click on **Next >**. The second step of the Lookup Wizard is displayed.

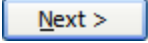


12. The **AcademicPlans** table is selected.

13. Click on . The next step of the Wizard is displayed.

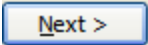


14. Click on the move all button . The Available Fields are moved to the Selected Fields column.

15. Click on . This step asked for the sort order for the list

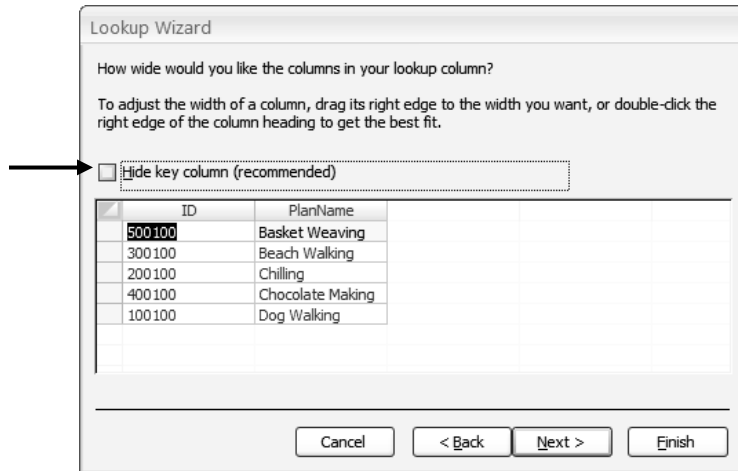
16. Select **PlanName** from the list at the first down arrow.



17. Click on .

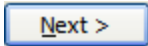
This step asks you to decide what columns will display in the list and how wide the columns should be.

18. **Unmark** the Hide key column checkbox if the primary key field should display.



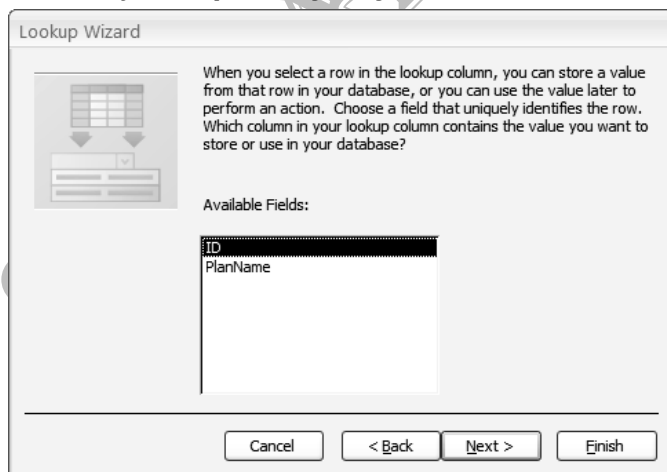
19. To change the width of any column:

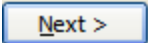
- Place the mouse pointer in the header on the column's right border.
- When the pointer is a double-headed arrow, drag to resize the column.

20. Click on .

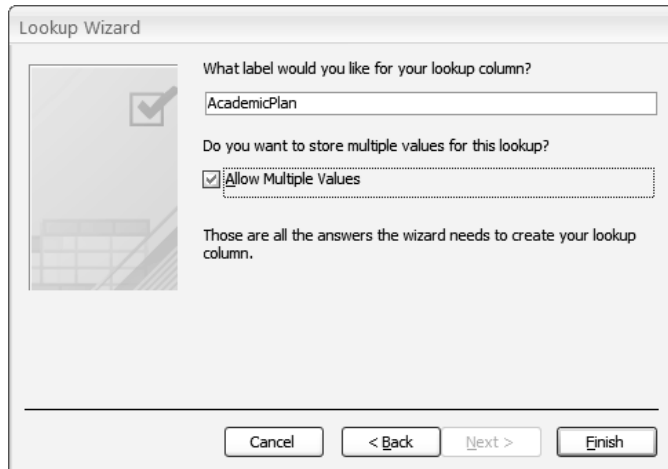
In this step, select the data to be stored in the table.

Generally, the **primary key** data is stored in the destination table.



21. Click on .

In this step, the label for the list and the option of “Allow Multiple Values” are available.

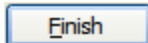


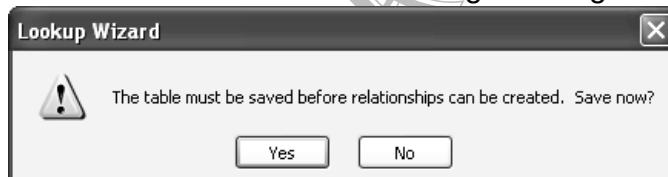
22. Type a label for the lookup column or accept the default fieldname.

In class, type **AcademicPlan**.

23. Mark or unmark the Allow Multiple Values checkbox, as appropriate.

In class, click to **mark the checkbox**.

24. Click on . The following message is displayed.



25. Click on **Yes**. The table is displayed.



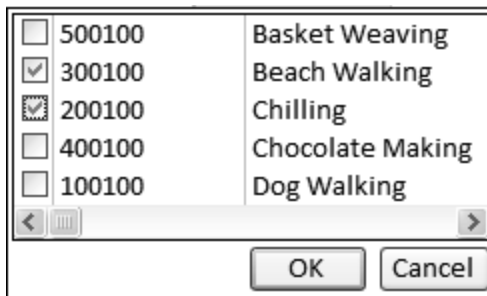
26. On the  tab in the field properties of this field, make the following revisions.

Column Widths	1";1.5"
List Rows	16
List Width	2.5"

27. **Save** the changes.

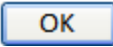
28. Use the View tool to switch to the **Datasheet view**.

29. In the first record, click on the down arrow for the Academic Plan field.

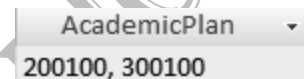


☐	500100	Basket Weaving
☒	300100	Beach Walking
☒	200100	Chilling
☐	400100	Chocolate Making
☐	100100	Dog Walking

OK Cancel

30. Select two values and click on 

31. The field values are shown with a comma as the separator.



AcademicPlan ▼

200100, 300100

Filter

Discussion

If you would like to see only the records that meet certain criteria, you can filter the records. When a filter is in place, the records that meet the criteria are shown and the other records are hidden from view.

In the Datasheet view of a table, the filter down arrows  are available to the right of a field name.

EmplID	LName	FName	CampusPhone	Email
--------	-------	-------	-------------	-------

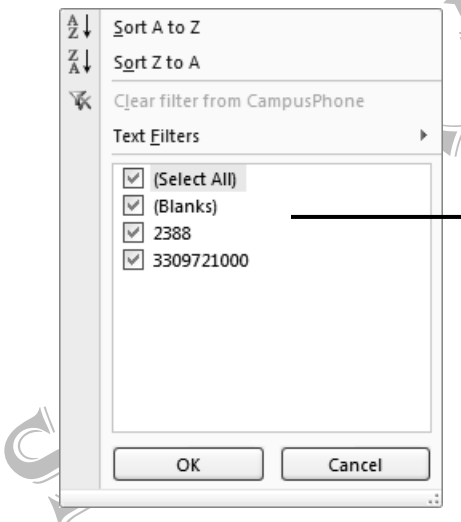
If you use the filter features in Excel, you will notice that the filter features in Access 2007 are similar.

Steps

1. Open the table in the Datasheet view.

In class, double click on **ContactStudent**, if it is not open.

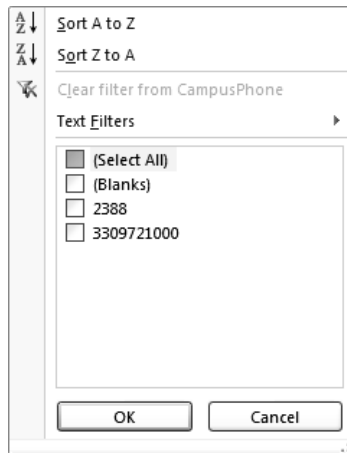
2. Click on the filter down arrow for CampusPhone. A list of choices is displayed.



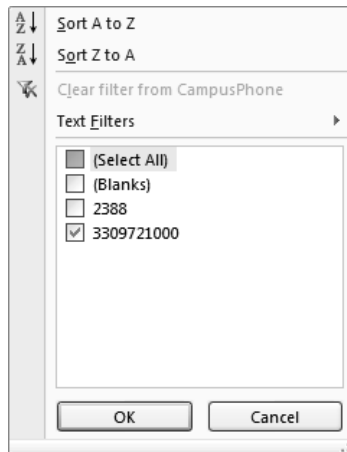
This is a list of the unique field values in this column.

3. To select specific field values for the records that you want to see:

- Unmark  (Select All) .



- Click to mark the field values that you want to see. Mark as many values as you want.



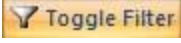
- Click on OK. Only the records that meet your criteria (selection) are displayed.

EmplID	LName	FName	CampusPhone	Email
2000	Wu	Henry	(330) 972-1000	wuhen@uakro
3000001	Miller	Bernice	(330) 972-1000	millber@uakrc
1234567	Jones	Hermione	(330) 972-1000	herm@uakron.
2356	Pietrowski	Joseph	(330) 972-1000	pietrow@uakr

Note that the filter icon  is displayed to the right of the field name. This indicates that all the records are not displayed.

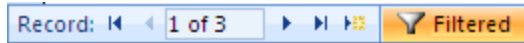
Note that  is displayed with the Navigation tools at the bottom of the window. This also indicates that all the records are not displayed.

4. To remove the filter and display all the records:

- Click on the Home tab and click on .

OR

- Click on the  button next to the Navigation tools.



5. To remove the filter for only one field:

- Click on the filter down arrow for the field with the filter in place.
- Mark Select All.
- Click OK.

Tip – Finding Records



The familiar Find tool  is available on the Home tab of the Ribbon. The new Search tool is available at the bottom of the Datasheet window.



The new Search tool can be used to find the first instance of a field value.

Click on the word Search and begin to type. Access searches all the records in the table forward from the active field value.

Use the Find tool to locate more than one instance.

Copy, Save and Delete Objects



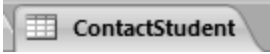
To save an object:

- Click on the Save tool  in the Quick Access toolbar .

OR

- Click the Office Button .
- Select Save.

OR

- Right-click the table's tab .
- Select Save from the shortcut menu.

OR

- Press CTRL + S.

To copy an object in the existing database:

- In the Navigation Pane, right-click on the object's name.
- Select Copy from the shortcut menu.
- Right-click in a blank area of the Navigation pane.
- Select Paste from the shortcut menu.
- If prompted for a name or to select an option, make the appropriate choices.

To delete an object in the existing database:

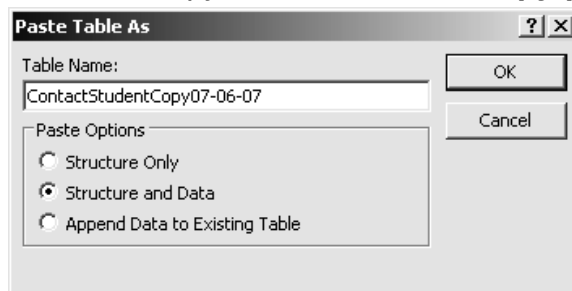
- In the Navigation Pane, right-click on the object's name.
- Select Delete from the shortcut menu.
- When prompted to confirm the deletion, click on Yes.

Exercise 3 - Copy and Delete an Object

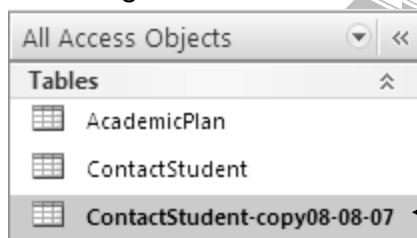


1. In the Navigation pane, right click on the **ContactStudent** table.
2. Select **Copy** from the shortcut menu.
3. Right-click in a blank area of the Navigation pane.
4. Select **Paste** from the shortcut menu.
5. When prompted:

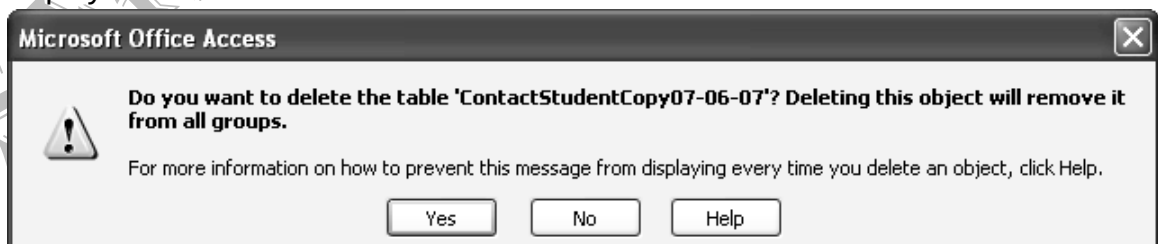
- Name the copy **ContactStudent-copy (today's date)**.



- Select the option to copy the **Structure and Data**.
- Click on OK. The table is copied to the database and its name is displayed in the Navigation Pane.



6. Right-click on the table name **ContactStudentCopy(date)**.
7. Select **Delete** from the shortcut menu. The confirm deletion message is displayed.



8. Confirm the deletion by clicking on Yes. The table is removed from the database.

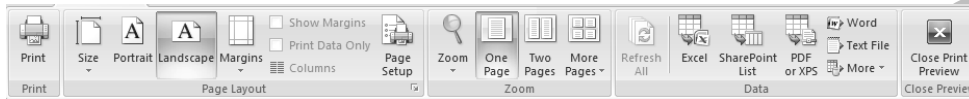
Printing



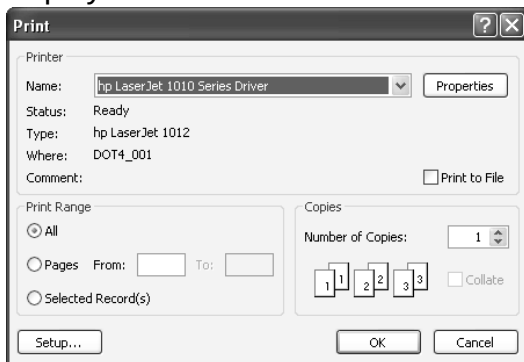
Print an open object

To print an open object, such as a table or query results:

1. Click on the Office Button.
2. Select Print, and then Print Preview. The Print Preview tab of the Ribbon is displayed.



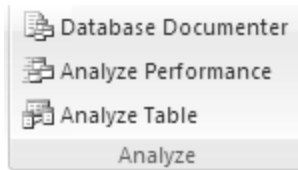
3. In the Print Preview tab, use the tools to make changes to the margins, paper orientation and/or page setup, as required.
4. When the print preview is what you want, click on . The Print dialog box will display.



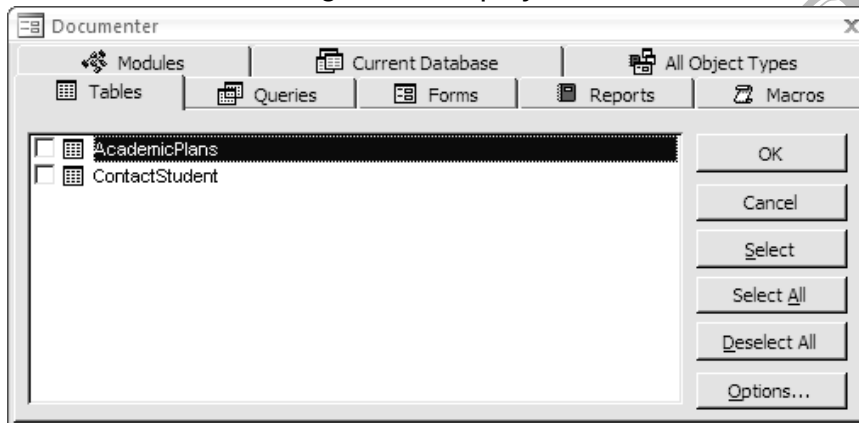
5. Make any selections in the Print dialog box and click on OK. The object will print.
6. Click on .

Print Table and Other Object Definitions

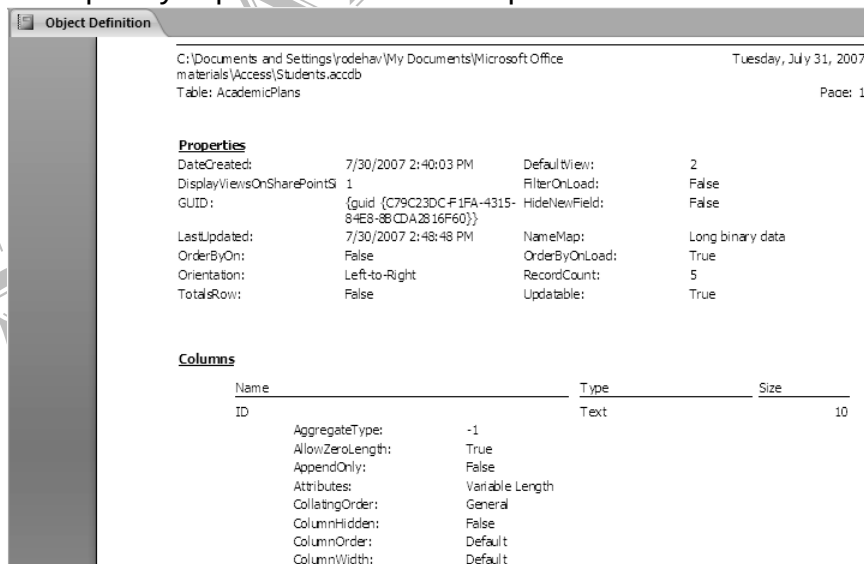
1. Click on the **Database Tools** tab.
2. In the Analyze group, select Database Documenter.



The Documenter dialog box is displayed.



3. Select the tab for the object to be documented.
4. Mark the objects to be documented.
5. Click OK.
6. A temporary report is created and opens in Print Preview.



7. Use the Print Preview tab to change print options, print the report and close the view.

Table Relationships

Discussion

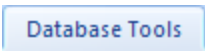
A relationship between tables can be established so that Access automatically will relate data between two tables. The primary key field in one table is related to a field that holds the same data in another table.

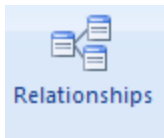
An example with which you are familiar is the EmplID field or SSN (Social Security Number) field that holds data that uniquely identifies each individual. The basic procedures for creating table relationships remain the same as in earlier versions of Access.

Steps

1. Open the database in which to create table relationship.

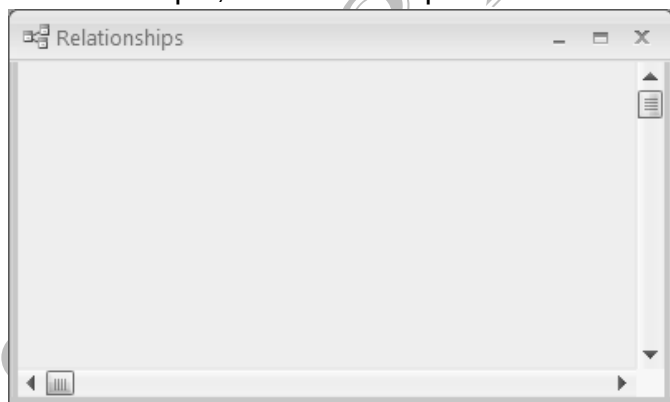
In class, open **Committees2007**.

2. Click on the  tab.

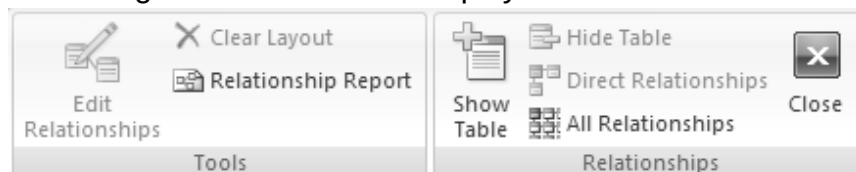
3. Select the  tool.

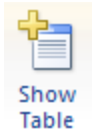
4. The Relationships window is displayed.

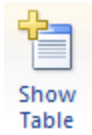
In this example, no relationships have been created so the window is empty.

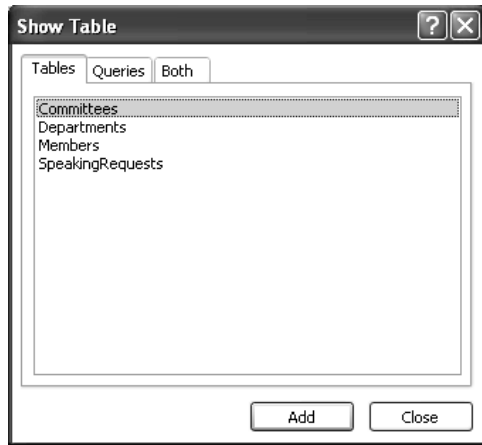


5. The Design contextual tab is displayed.





6. Click on the  tool. The Show Table dialog box is displayed.



7. Select the first table in the relationship. This is the table with the field as the primary key.

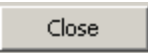
In class, select **Departments**.

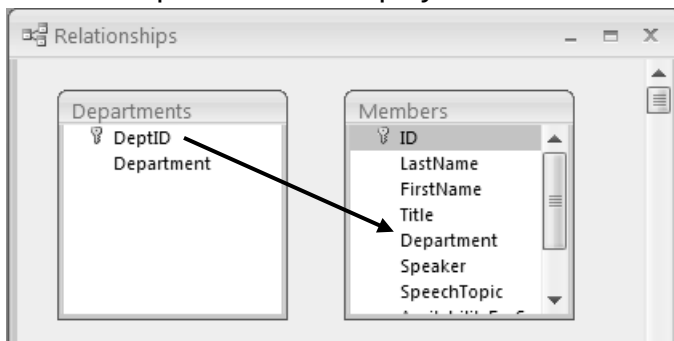
8. Click on **Add**. The table is added to the Relationship window.

9. Select the second table in the relationship.

In class, select **Members**.

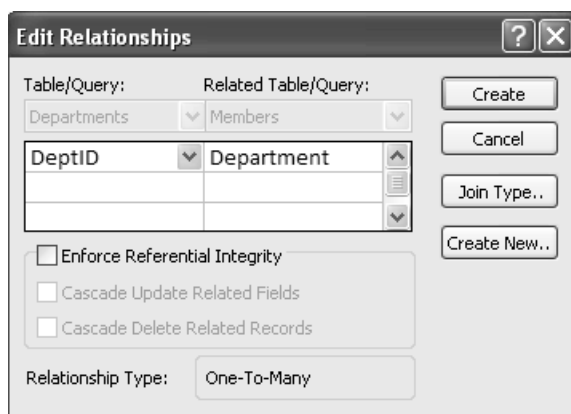
10. Click on **Add**. The table is added to the Relationship window.

11. Click on  in the Show Table window dialog box to close it. The Relationship window is displayed.



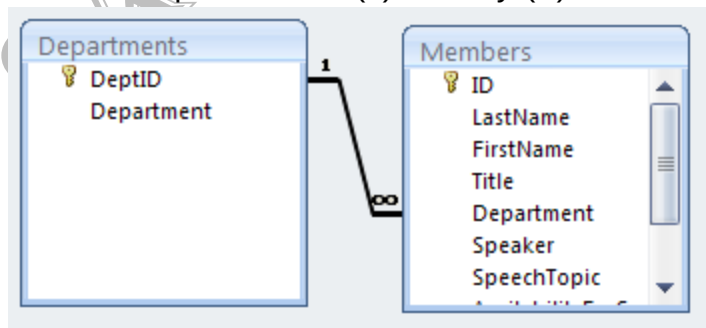
12. Use the mouse to drag from the primary key field in the first table to drop on the related field in the second table.

In class, drag from the **DeptID** field in the Departments table to the **Department** field in the Members table. The Edit Relationships box is displayed.



13. Optional: Mark the box to Enforce Referential Integrity.
14. Click on **Create**. The join line is drawn to show the relationship.

In this example, the one (1) to many (∞) relationship is shown.



15. Click on the  **Relationship Report** tool in the Design tab to create a report that can be printed and saved.

Validation Rules

Discussion

When designing a table, validation rules can be used to control how data is entered in a field. If a data entry does not meet the rule, an error message can be displayed. You can establish the text of that error message.

If you create a validation rule for a field, Access does not normally allow a Null value to be stored in the field. If you want to allow a Null value, add "IS Null" to the validation rules, as in "<>8 Or is Null" and make sure the Required field property is set to No.

A validation rule is created in the Design view of a table. The rule is a field property, so it is entered in the lower pane of the design window.

Field Name	Data Type
ReqID	AutoNumber
RequesterName	Text
PlusFour	Text
Phone	Text
Date	Date/Time
Time	Date/Time
SpeechLocation	Text
SpeakerID	Text

Field Properties	
General Lookup	
Format	Short Date
Input Mask	
Caption	
Default Value	
Validation Rule	>=#1/1/2007#
Validation Text	
Required	No
Indexed	No
IME Mode	No Control
IME Sentence Mode	None
Smart Tags	
Text Align	General
Show Date Picker	For dates

Validation rules in Access 2007 are created as they were in earlier versions of Access.

Steps



1. In the Design view of the table, click in the field for which a validation rule will be created.
2. In the Field Properties area of the window, click in the Validation Rule property.
3. Type the rule or click on the Expression Builder tool  to help. (Remember that the Expression Builder tool is displayed after you click in the edit box for the property.)
4. In the Validation Text property, type the text that is displayed if a person enters data that does not meet the rule's requirement.

Validation Rule	>=#1/2/2007#
Validation Text	Enter a date after 01/01/07.

5. Example:
6. If there is already data in the table, Access displays a message to ask if you want to check the existing data for errors.
7. If you indicate yes, Access checks the data to see if it follows the rule. If any data breaks the rule, a message displays to advise you. If some of the data breaks the rule, but you want to keep the rule, click on Yes.

Lesson 4: Forms

Discussion

Forms frequently are used as a way of entering data into the tables or looking up records in the tables. Forms make data from a table or a query easier to understand and interpret by presenting it in a visually-appealing manner.

A basic form can be created quickly by using the tools in the Forms group on the Create tab of the Ribbon. Access 2007 has a new form available called the Split Form. In a Split Form, the top of the form displays one record at a time in a standard Access format. The bottom of the form displays the data in a Datasheet type of view.

There is a new view available for Forms - Layout view. In Layout view, the form is displayed with the data and you also can make changes to its design. Layout view gives the benefit of the Form view (where the data is displayed and entered) and some of the benefits of the Design view (where the form is created and revised). Some design tasks cannot be performed in the Layout view. Design view is required for those tasks. Design view still can be used to perform all design tasks, including those that can be accomplished in Layout view.

Lesson Topics

In this lesson, you will:

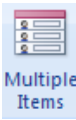
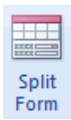
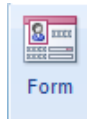
- ▶ Create a basic Form.
- ▶ Edit a Form
- ▶ Use the Form wizard
- ▶ Filter data in a Form
- ▶ Print a form

Create a Basic Form



In class, open the Committees2007 database, if it is not open.

1. Select the table or query on which the form is to be based.
In class, select the **Members** table.
2. Click on the Create tab of the Ribbon
3. Depending on the type of form that is to be created,



click on the **Form** OR **Split Form** OR **Multiple Items** tool.

In class, try each tool.

Note: When creating a form for a table or query, choose the type of form that presents the data in the best way. For one table, the Form type may present the data well, while the Split Form type may be best for another table.

4. When a form is created, it defaults to the new view - Layout view.

5. The form created with the **Form tool** is:

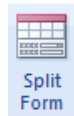
Tip – Edit a Form

When a new form is created and is not yet saved, the form initially is displayed in the Layout view. The

Format **Arrange**

contextual tabs are available in this view. These tabs contain tools to edit the form.

If printing is important, note that records in a Split Form print in Datasheet format, not in Form format.



The form created with the Split Form tool is:

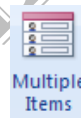
Members

ID: 1
 LastName: Smith
 FirstName: Adele
 Title: Staff
 Department: HIKE
 Speaker: ☐

SpeechTopic:
 AvailabilityForSpeaking:
 Committee: PROCON
 Phone: (330) 972-0101

ID	LastName	FirstName	Title	Department	Speaker	SpeechTopic	Availability
1	Smith	Adele	Staff	HIKE	☐		
2	Rodehaver	Pam	Software Train	HIKE	☒	Technology in	Friday 8:00 A.M.
3	Smith	Mark	Professor	MCHENG	☐		
4	Jones	Robert	Staff	HIKE	☒	Hiking in Summ	Wednesday 3:00
5	Harris	Juan	Associate Profr	CHEM	☐		
6	Gonzales-Smith	Mary	Instructor	MCHENG	☐		
7	Wyszynski	Sara	Instructor	CHEM	☐		
8	Lee	James	Staff	CHEM	☒	A Brief History	Tuesday 12:00

Record: 1 of 9 | No Filter | Search



The form created with the Multiple Items tool is a list with each record in a row.

Members1

ID	LastName	FirstName	Title	Department	Speaker	SpeechTopic
1	Smith	Adele	Staff	HIKE	☐	
2	Rodehaver	Pam	Software Trainer	HIKE	☒	Technology in Microsoft Office Instruction
3	Smith	Mark	Professor	MCHENG	☐	
4	Jones	Robert	Staff	HIKE	☒	Hiking in Summer of 5 Suggested
5	Harris	Juan	Associate Professor	CHEM	☐	
6	Gonzales-Smith	Mary	Instructor	MCHENG	☐	
7	Wyszynski	Sara	Instructor	CHEM	☐	

Record: 1 of 9 | No Filter | Search

Edit a Form

Discussion

A form may be edited in one of two views:

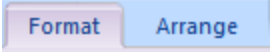
- Layout view
- Design view



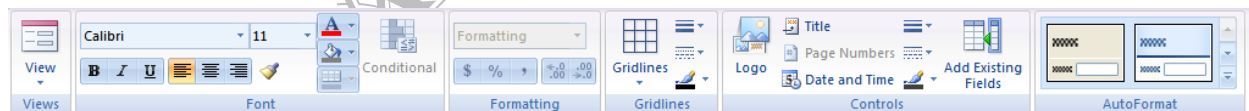
Use the  tool on the Home tab of the Ribbon to switch between the three main views of a form:



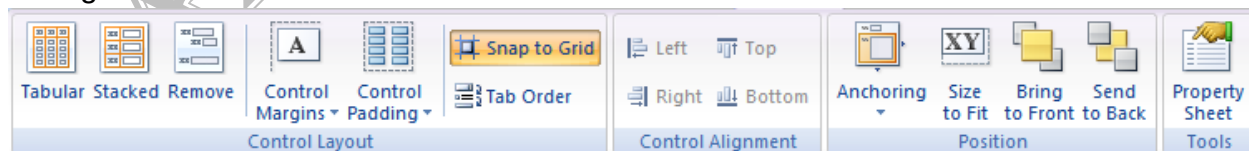
A form also may have PivotTable and PivotChart views available.

When a form is created using the Form Wizard, the form initially is displayed in the new view, Layout view. In this view, many of the design changes that affect the appearance of the form can be made. In Layout view, the  contextual tabs are available on the Ribbon. These tabs display tools to assist with Form design changes.

Format tab



Arrange tab



After a form is saved and closed, the form is displayed in Form view, when it is reopened. The Format and Arrange tabs are not displayed. To return to Layout view,

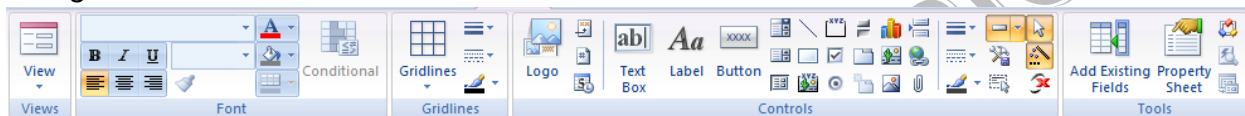


click on the  tool on the Home tab and select Layout view.

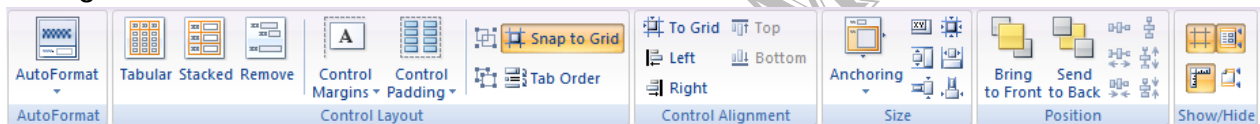
Not all design changes can be made in Layout view. All design changes can be made in Design view. When Design view is selected from the View tool on the Home tab, the

 tabs are displayed on the Ribbon.

Design tab



Arrange tab



Steps

1. A form may be edited, in part, in the Layout view. In class, switch to the first form that you created.

ID:	1
LastName:	Smith
FirstName:	Adele
Title:	Staff
Department:	HIKE
Speaker:	☐

Tip – Disable Layout View

If you do not want anyone to edit the form unless it is in Design view, disable the Layout View. The instructions are on page 99.

2. Look at the left side of the status bar, which is at the bottom of your window. Confirm that the form is in **Layout View**. If it is not, click on the View tool on the Home tab of the Ribbon to select Layout view.

- To move a field, place the mouse over it. With the mouse as a four-pronged arrow, drag the field to a new location.

In class, click on the FirstName field and drag it to above LastName. Drop FirstName when the line displayed is above the LastName field.

- Click on the Save tool in the Quick Access toolbar. The Save As dialog box is displayed.

- Click on OK to accept the name Members.
- Close and reopen the form.
- Try to move the LastName field. It cannot be moved in Form view.
- Note that after the Form is saved, the Format and Layout contextual tabs of the Ribbon are not available.
- Click on the **Members1** tab. This is a Split Form that you created and have not saved. It is in Layout view.
- Click on the LastName label to select it.
- Hold down the Shift key and click on the FirstName label to select it, also.
- Select the **Format** tab in the Ribbon. Make these changes.
 - Click on the Font Size tool **11** and select 14 pt.
 - Click on the Bold tool **B**.
 - Click on the down arrow of the Fill tool and select a shade of green.



13. Save the changes to the form. Name it Members1.

14. To edit a form after it is closed and saved:

- Click on the Home tab of the Ribbon.



- Select the  tool. A list is displayed.
- Select the Layout view or Design view from the list.

In class, switch to Design view.

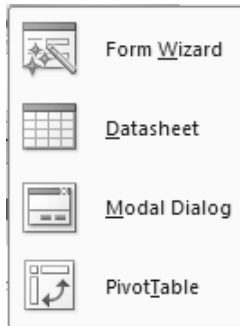
15. Use the View tool on the Home tab of the ribbon to switch to Form view.

Use the Form Wizard

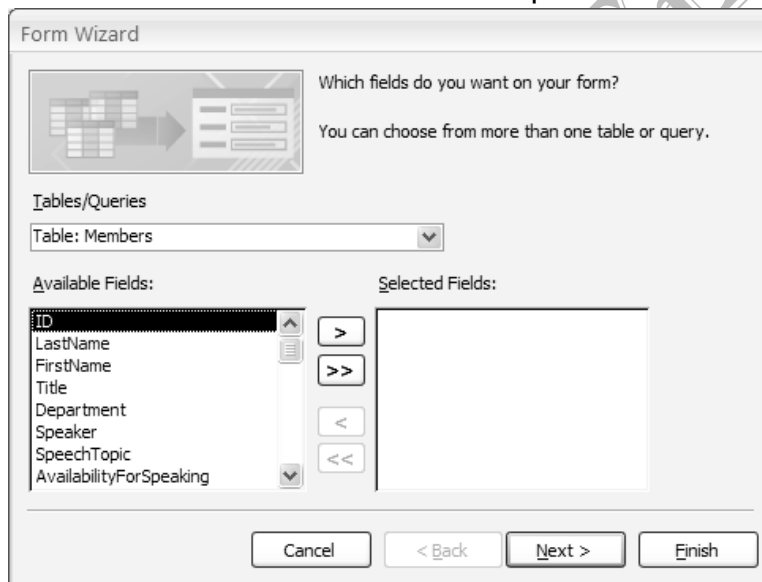


The familiar Form Wizard still is available. To use it:

1. Click once on the Table or Query that will be the source for the Form.
2. On the **Create** tab, in the Forms group, select the **More Forms** tool. A list of options is displayed.



3. Select **Form Wizard**. The first step of the Form Wizard is displayed.



4. Follow the instructions to move through the steps to create a form.
5. Switch to the Design view to edit the form.

Filter



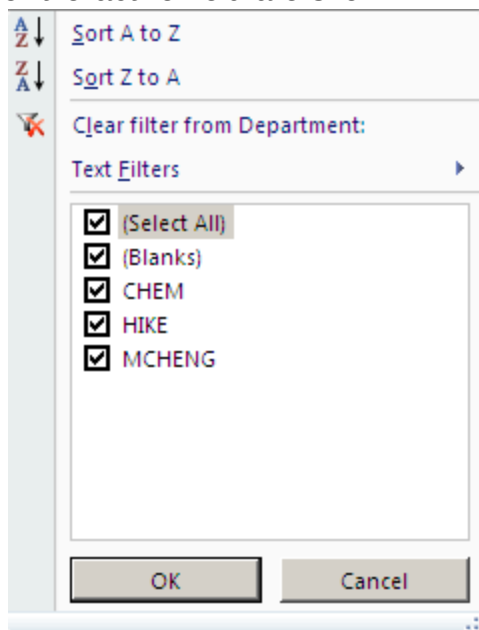
If you would like to see only the records that meet certain criteria, you can filter the records in Form view. When a filter is in place, the records that meet the criteria are shown and the other records are hidden from view.

If you use the filter features in Excel, you will notice that the filter features in Access 2007 are similar.

1. In the Form view, click in the field to be used for the criterion.
In class, click in the **Department** field.



2. Click on the Filter tool on the Home tab of the Ribbon. The Filter list options for the active field are shown.

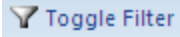


3. To select specific field values for the records that you want to see:
4. Unmark ☒ (Select All).
5. Click to mark the field values that you want to see. Mark as many values as you want.

In class, mark **Hike**.

6. Click on OK.
7. Use the Navigation tools in the bottom left corner of the window to view the records for the Hiking (HIKE) Department.

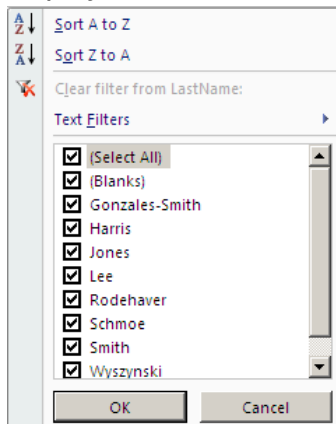


8. To remove the filter so all the records can be viewed, click on  in the Home tab of the Ribbon.

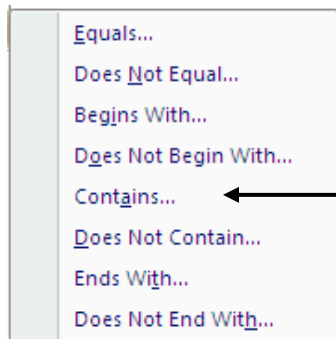
OR

Click on the  button next to the Navigation tools.

9. Click in the **LastName** field. It will be used for the next filter.
10. Click on the Filter tool in the Home tab of the Ribbon. The list of Filter options is displayed.

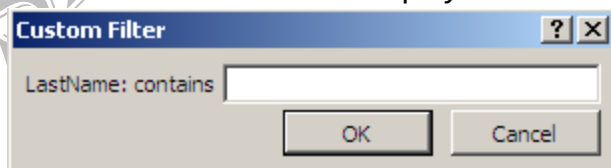


11. Point to Text Filters. The list of options is displayed.



12. Select the appropriate option for the criterion.
In class, the criterion is LastName **contains** Smith. Select contains.

The Custom Filter box is displayed.



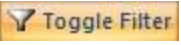
13. Enter the text to complete the criterion.
In class, type **Smith**.
14. Click on OK.

15. Only the records that meet your criterion are displayed.

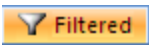
Use the Navigation tools in the bottom left corner of the window to view the records for Members whose last name contains Smith.



16. To remove the filter and display all the records:

- Click on the Home tab and click on .

OR

- Click on the  button next to the Navigation tools.

Print

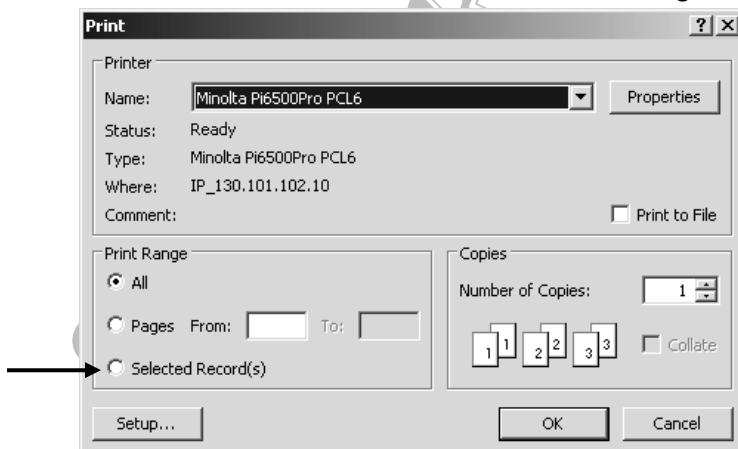


To print all the forms:

- Click on the Office Button.
- Select Print and then Quick Print.

To print a single form:

- Navigate to the individual form.
- Click on the Office Button.
- Select Print and then Print. The Print dialog box is displayed.



- Click to choose Selected Record(s).
- Click on OK.

Lesson 5: Queries

Lesson Topics

In this lesson, you will:

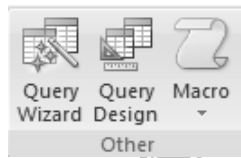
- ▶ Create a Query in Design view
- ▶ Review object dependencies
- ▶ Export data to Excel
- ▶ Export data to Word
- ▶ Create a PDF file

Create a Query



To create a new query:

- Select the **Create** tab of the Ribbon.



- In the Other group, select either:
Query Wizard (to use the Crosstab, Simple, Find Duplicates or Find Unmatched Wizard)

OR

Query Design (to create the query in Design view).

To run a query:

- In the Navigation pane, click on the down arrow in the title bar to make selections.
- Select the Object Type category, if necessary.
- Select the Queries in the Filter by Group list.
- Double click on the name of the Query to run. The results are displayed.

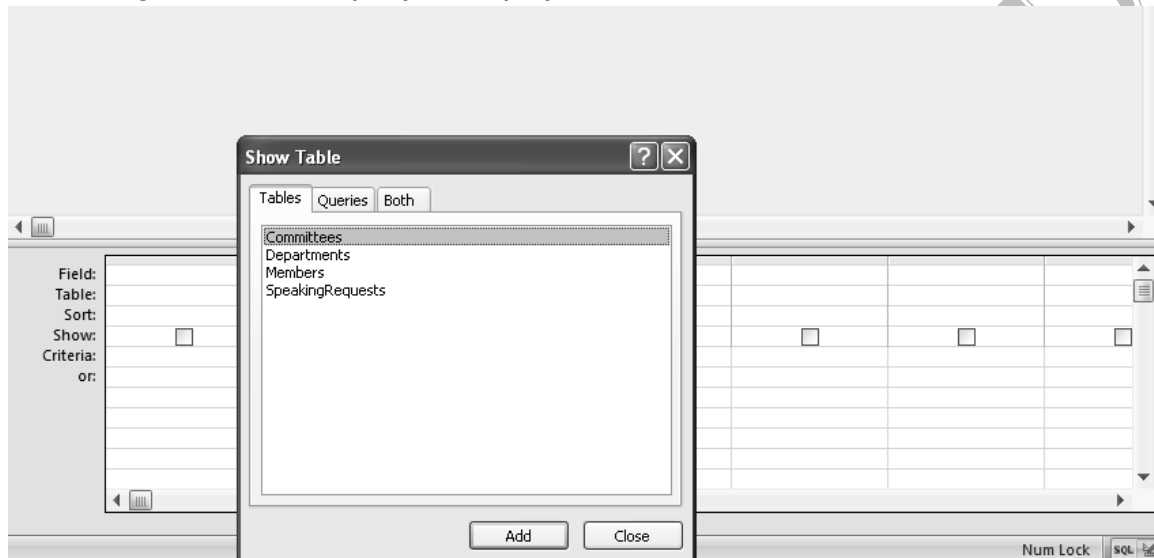
To print the query results:

- With the query results displayed, click on the Office Button.
- Select Print and then Print. The Print dialog box is displayed.
- Make any changes to the Print options.
- Click OK.

Exercise 4 – Create a Query in Design View



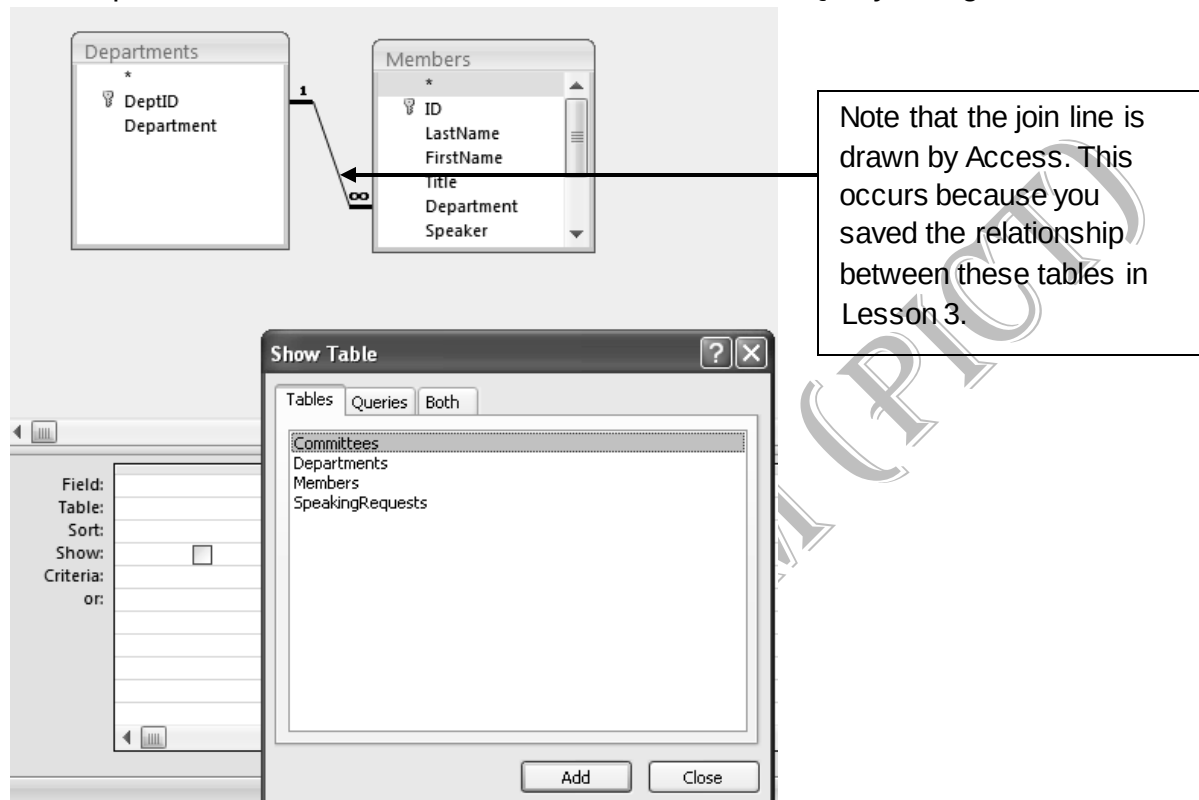
1. Click on the Create tab.
2. In the Other group, click on Query Design.
The Design view of the query is displayed.



3. In the Show Table box, click on **Departments** and then **Add**.

4. Click on **Members** and then **Add**.

The Departments and Members tables are added to the Query design.



5. Click on Close to close the Show Table box.
 6. From the Members table, select these fields:
LastName
FirstName

As in previous versions of Access, double click on a field or drag it to select it.

7. From the Departments table, select **Department**.

The Query grid looks like this:

Field:	LastName	FirstName	Department
Table:	Members	Members	Departments
Sort:			
Show:	☒	☒	☒
Criteria:			
or:			

8. In the Sort row, click in the LastName column and choose Ascending.

9. In the Sort row, click in the FirstName column and choose Ascending.
The Query grid looks like this:

Field:	LastName	FirstName	Department
Table:	Members	Members	Departments
Sort:	Ascending	Ascending	
Show:	☒	☒	☒
Criteria:			
or:			

10. Click on the Save tool  in the Quick Access toolbar.
11. Name the query **MembersDeptList-Q**.



Save As dialog box with Query Name: MembersDeptList-Q. Buttons: OK, Cancel.

12. Click on OK.



13. Click on the  tool on the Design tab of the Ribbon.
The query results are displayed.

LastName	FirstName	Department
Gonzales-Smith	Mary	Mechanical Engineering
Harris	Juan	Chemistry
Jones	Robert	Hiking Cross-Country
Lee	James	Chemistry
Rodehaver	Pam	Hiking Cross-Country
Schmoe	Joan	Chemistry
Smith	Adele	Hiking Cross-Country
Smith	Mark	Mechanical Engineering
Wyszynski	Sara	Chemistry
*		

14. Close the Query window by clicking on its X.
Note that when the mouse points to the X, there is a tool tip that says 'Close MembersDeptList-Q'.



Object Dependencies

Discussion

It is important to know if an object (source object or control object) is related to another object. Changes or deletion in the source object may adversely affect the related object.

To determine if an object has other objects that depend upon it, use the View Dependencies command.

In order for the View Dependencies command to work, the Track Name AutoCorrect Info box must be marked in Access options. It is suggested that the Perform name AutoCorrect box also be marked, although View Dependencies does not require it. Name AutoCorrect is the function that repairs references to renamed objects.

Set the AutoCorrect Options

1. To set the Track Name AutoCorrect option for the current database, click on the Office Button and select **Access Options**. The Access Options dialog is displayed.
2. In the left column of the dialog box, select **Current Database**.
3. Scroll down in the right column of the dialog box to view the Name AutoCorrect Options area.

Name AutoCorrect Options

- ☐ Track name AutoCorrect info
- ☐ Perform name AutoCorrect
- ☐ Log name AutoCorrect changes

4. Mark the box to **Track name AutoCorrect info**. This allows Access to store the information needed to correct naming errors.
5. If this message is displayed, click on OK.



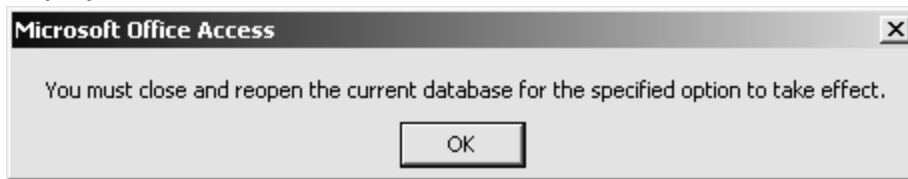
6. In order for the naming errors to be corrected, you also must mark the box for Perform name AutoCorrect.

Name AutoCorrect Options

- ☒ Track name AutoCorrect info
- ☒ Perform name AutoCorrect
- ☐ Log name AutoCorrect changes

7. OPTIONAL: If you would like to know what naming errors Access has found and corrected, mark the Log name AutoCorrect changes box. Access will create a table named AutoCorrect.log that may be viewed.

8. Click on OK. The dialog box is closed, there is a pause and the following message is displayed.



9. Use the Office Button to close the database. Then, reopen it.

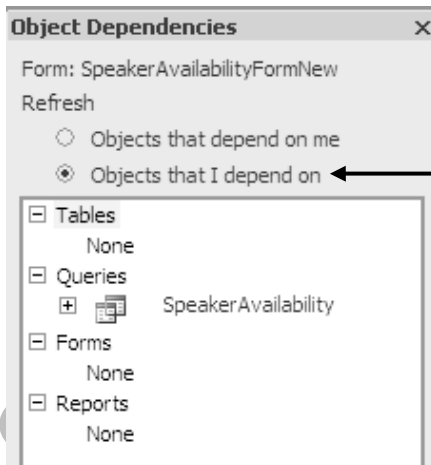
View Object Dependencies

1. The Track name AutoCorrect info box must be marked in the Current Database area of Access Options. Follow the previous steps to select that option.
2. In the Navigation Pane, click once on the object for which to view the dependencies.

In class, select the form named **Speaker Availability Form New**.

3. Click on the **Database Tools** tab.

4. Select  **Object Dependencies**. The Object Dependencies pane is displayed.

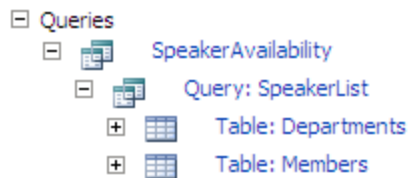


5. Select the option **Objects that I depend on**.
6. Note that the form depends on a query named Speaker Availability.
7. Click on the Expand tool .

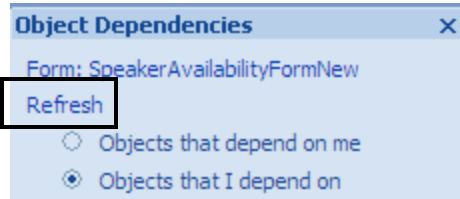


8. Note that the query Speaker Availability depends on the query SpeakerList.

9. Click on the Expand tool  .



10. Note that the query SpeakerList depends on the tables Departments and Members.
11. In the Navigation pane, click once on the form Members.
12. In the Object Dependencies pane, click on Refresh.



13. Note the dependencies for the Members form.
14. Close the Object Dependencies pane by clicking on its X.

Export Data

Discussion

Data may be exported in several different formats. Common exports are:

- To Excel in 2007 or 97-2003 format
- To Word as an RTF file
- As a PDF file that can be opened in Adobe Reader
- As a text file
- To another Access database

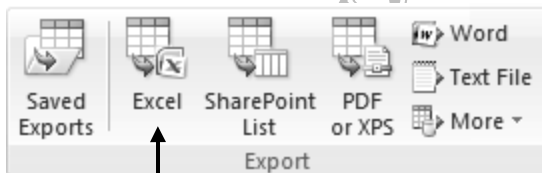
In this part of the lesson, you examine the exports to Excel, Word and to a PDF file.

Export Data to Excel

1. Click to select the data to be exported. You may click on the object's name in the Navigation pane or click on the open object, such as a Table or Query in the datasheet view.

In class, select the **MembersDeptList-Q**.

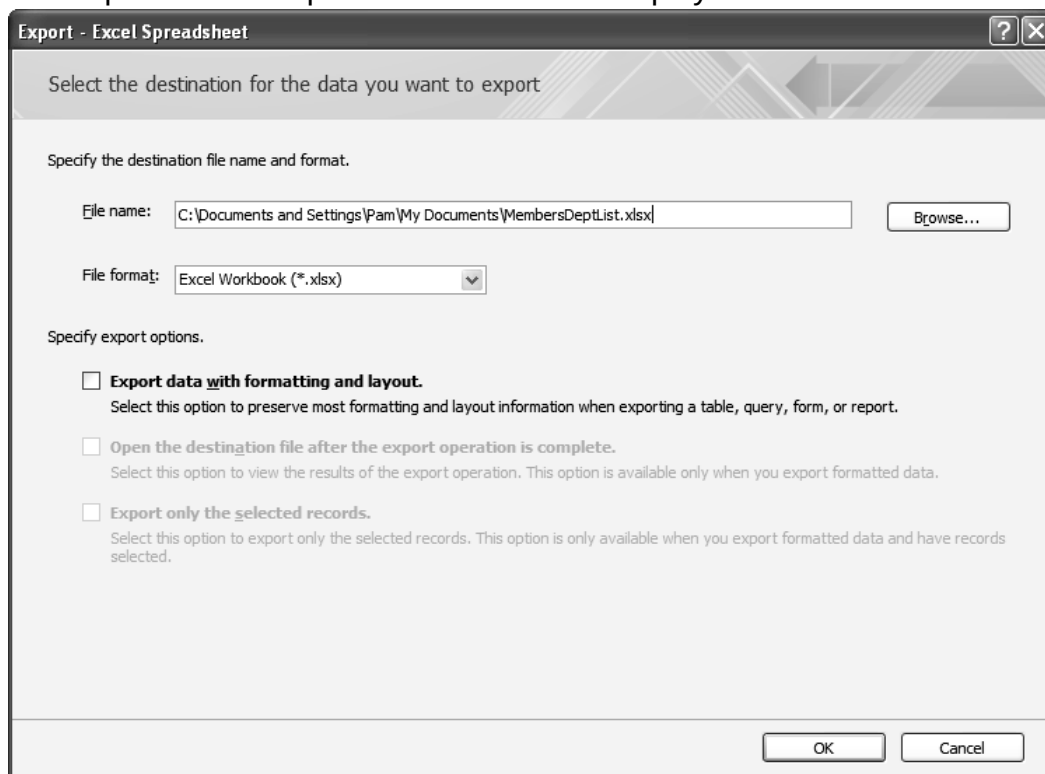
2. Click on the **External Data** tab of the Ribbon.
3. In the Export group, click on the Excel tool.



Tip – Export Data

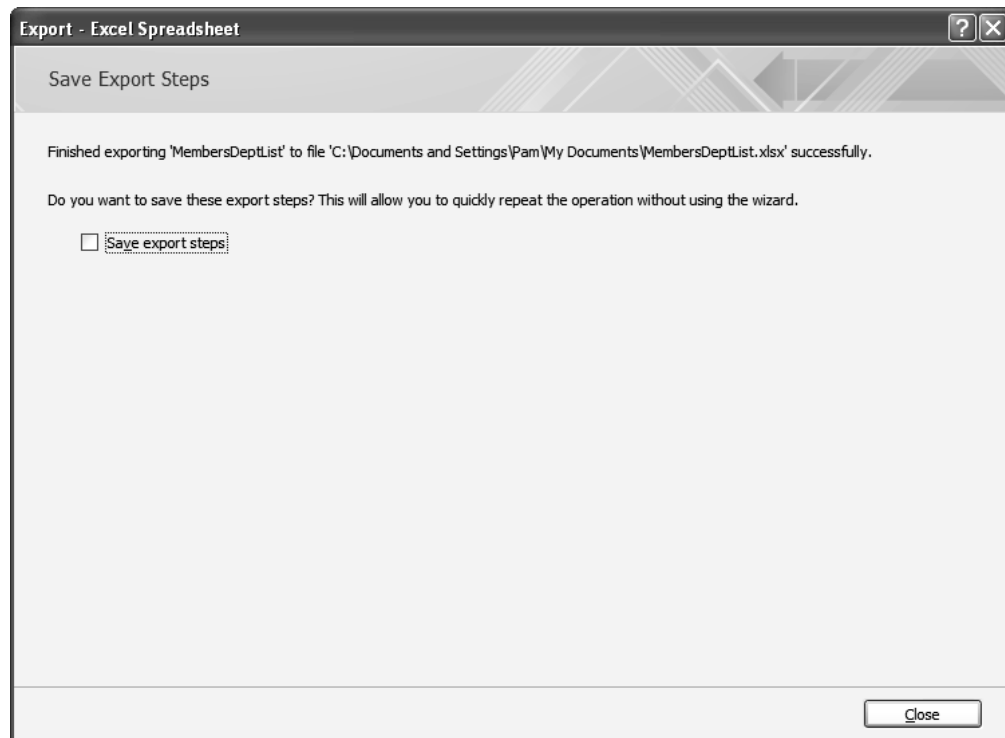
Another method is available to begin the Export process. Right click on the object's name in the Navigation Pane. From the shortcut menu that appears, select Export and the format.

The Export – Excel Spreadsheet wizard is displayed.

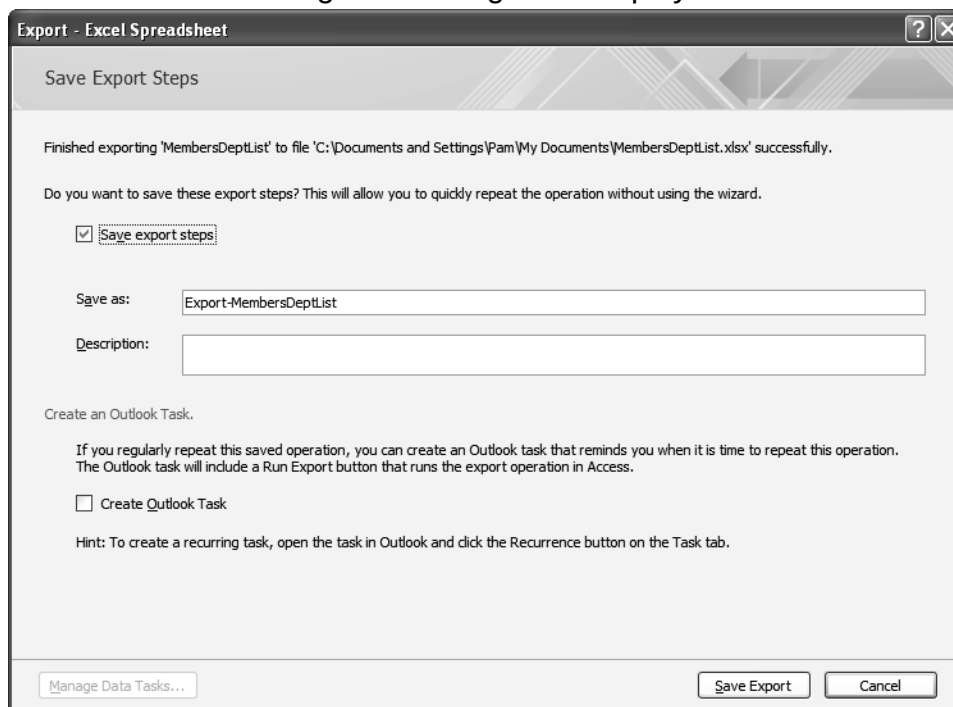


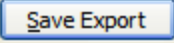
4. In the File name edit box, make any changes or use the Browse button to navigate to another drive or folder.
5. In the File format list box, select the format for the exported file.
6. In the Specify export options, mark the appropriate box(es). Some of the options may be unavailable, because they are not valid options for this export.

7. Click on OK. A second box is displayed. If the export was successful, the message is:



8. OPTIONAL: If this export is done frequently, click on the Save export steps checkbox. The message box changes to display additional choices.



- In the Save As edit box, type a name for the Export process. When you want to export this same object in the future, you will select this name from a list.
- OPTIONAL: In the Description box, type a short description of the Export process that is being saved.
- OPTIONAL: Mark the Create Outlook Task box if you would like a reminder to run this Export process at a particular date/time in the future.
- Click on .

Note: To run a saved export process in the future:

- Click on the External Data tab of the Ribbon.

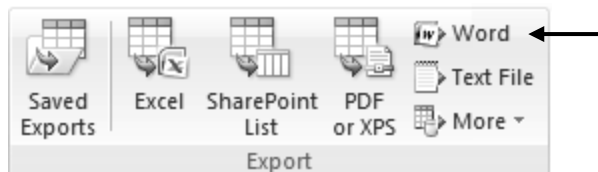


- Click on the  tool. A box is displayed with a list of the saved processes by name, file name for export and description.
- Change any of the items by double clicking on the item. Unless the new exported data should overwrite the previously exported data, double click on the file name for the export and give the new file a different name.
- Click on Run.
- Respond to any messages from Access.
- Close the Managed Data Tasks box.

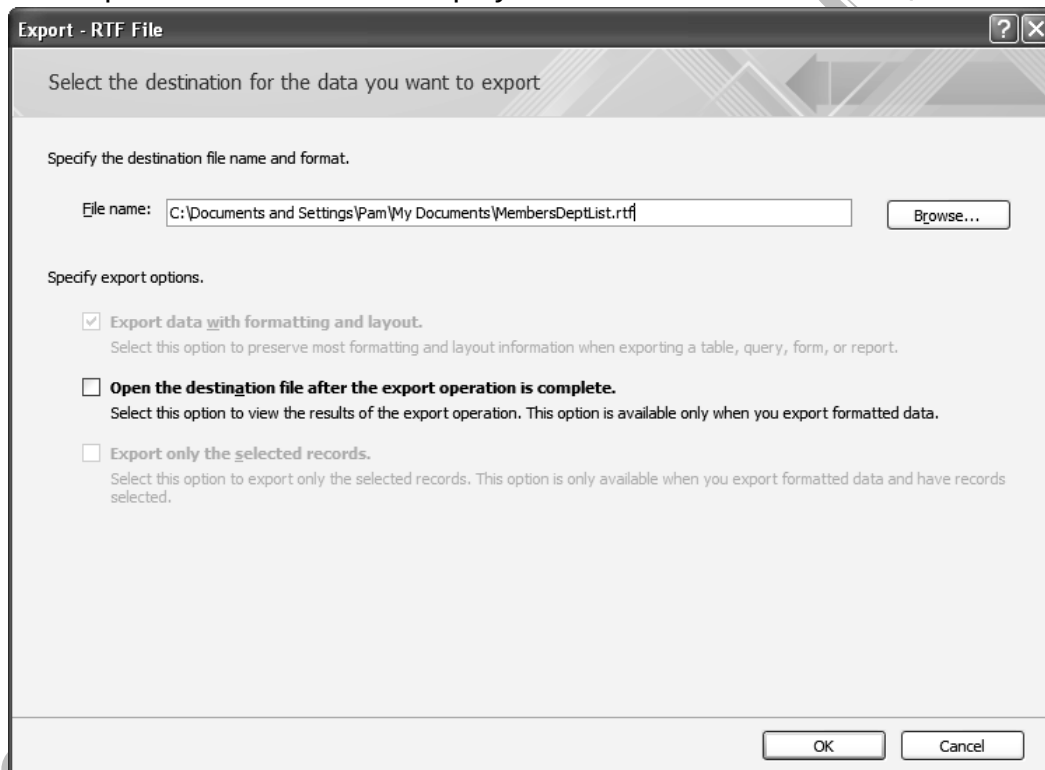
Export Data to Word



1. Click to select the data to be exported. You may click on the object's name in the Navigation pane or click on the open object, such as a Table or Query in the datasheet view.
2. Click on the **External Data** tab of the Ribbon.
3. In the Export group, click on the Word tool.

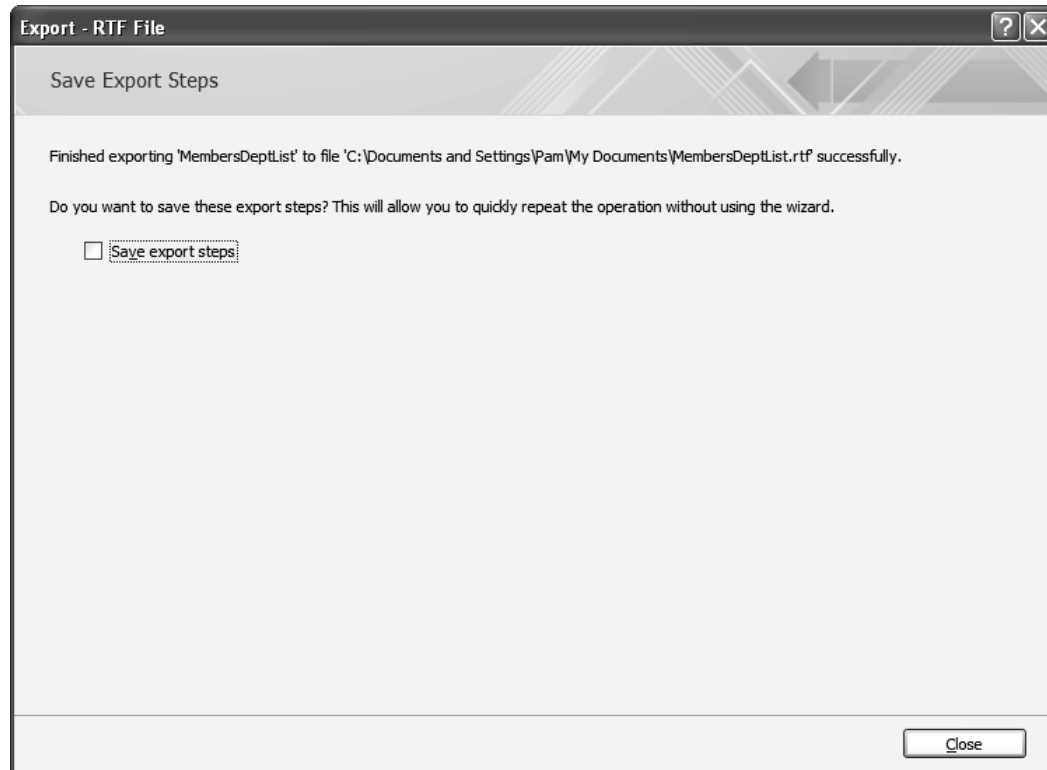


The Export – RTF file box is displayed. RTF stands for rich text file.



4. In the File name edit box, make any changes or use the Browse button to navigate to another drive or folder.
5. In the Specify export options, mark the appropriate box(es). Some of the options may be unavailable, because they are not valid options for this export.

6. Click on OK. A second box is displayed. If the export was successful, the message is:



7. OPTIONAL: If this export is done frequently, click on the Save export steps checkbox. The message box changes to display additional choices.

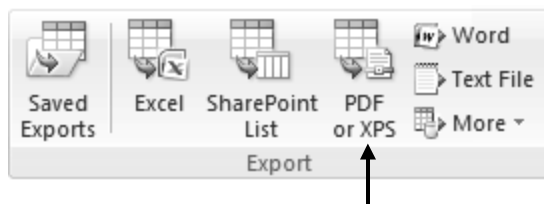
Refer to step 8 on page 75 for Saving an export process.

8. Click on Close to exit the wizard.

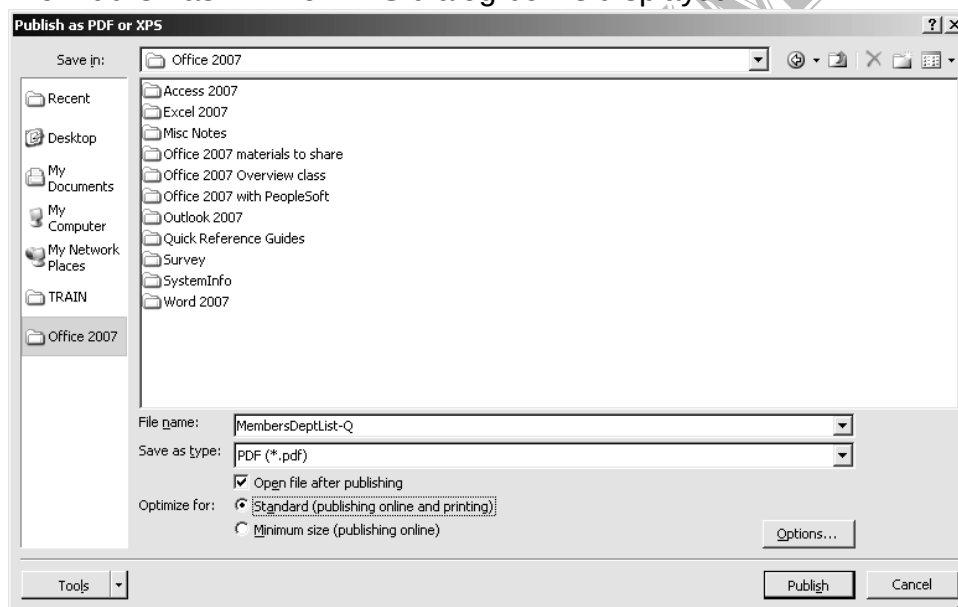
Create a PDF File



1. Click to select the data to be exported. You may click on the object's name in the Navigation pane or click on the open object, such as a Table or Query in the datasheet view.
2. Click on the **External Data** tab of the Ribbon.
3. In the Export group, click on the PDF or XPS tool.



The Publish as PDF or XPS dialog box is displayed.



4. Navigate to the drive and folder in which to save the PDF file.
5. Enter a name for the PDF file.
6. OPTIONAL: Mark the ☐ **Open file after publishing** if you want Adobe Reader to open and display the file.
7. Select an Optimize option. If you are in doubt, select the Standard option.
8. Click on Publish. The Export PDF box is displayed.
9. OPTIONAL: If this export is done frequently, click on the Save export steps checkbox. The message box changes to display additional choices.
10. Refer to step 8 on page 75 for saving an export process.

Lesson 6: Reports

A report object in Access contains the saved instructions for preparing a formatted copy of the data in a table or the results of a query. Reports can make data more useful through organization or summary.

Lesson Topics

- ▶ In this lesson, you will:
- ▶ Use the Report Wizard.
- ▶ Edit a Report
- ▶ E-mail a Report
- ▶ Use Report View

Use the Report Wizard

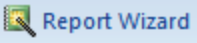


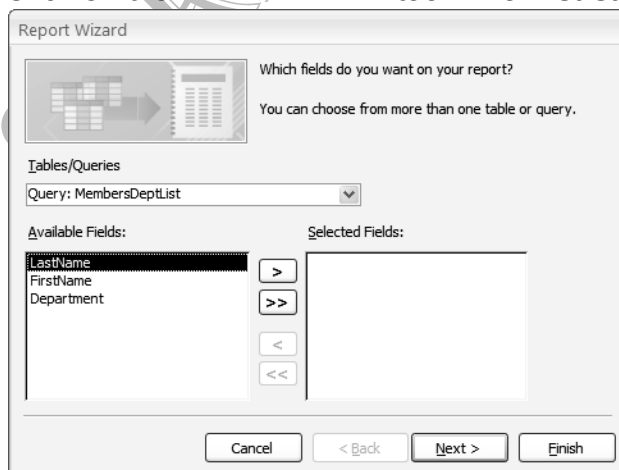
The Report Wizard can be helpful in the initial design of a report. It has the same steps as it did in Access 2003.

Upon completion of the wizard's process, you may choose to make changes in the arrangement of data and other controls, such as labels and lines, or to add controls.

1. In the Navigation pane, select the object that is the source for the report. This is usually a query, but may be a table.

In class, select the Query named **MembersDeptList-Q**.

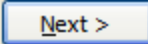
2. Click on the  tab of the Ribbon.
3. Click on the  tool. The first step of the Report Wizard is displayed.

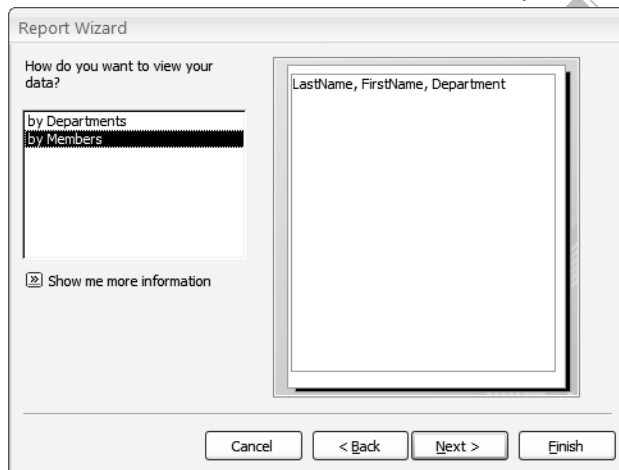


4. From the Available Fields box, select the fields for the report.
Select the fields in the order that they will be displayed in the report.

In class, select all the fields in their current order by clicking on the  tool.



5. Click on . The second step of the wizard is displayed.



This step of the wizard asks about the order/grouping for the data.

6. Select how the records should be grouped.

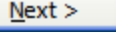
In class, click on **By Departments** to order and group the records by the department value.

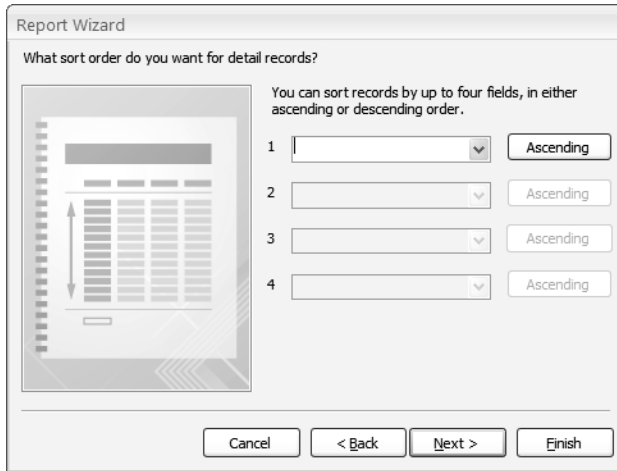


7. Click on **Next >**. The third step of the wizard is displayed.



This step of the wizard asks about any additional grouping for the data.

8. Select any additional grouping in this step. Click on . The next step of the wizard is displayed.



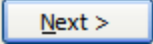
Tip - Grouping

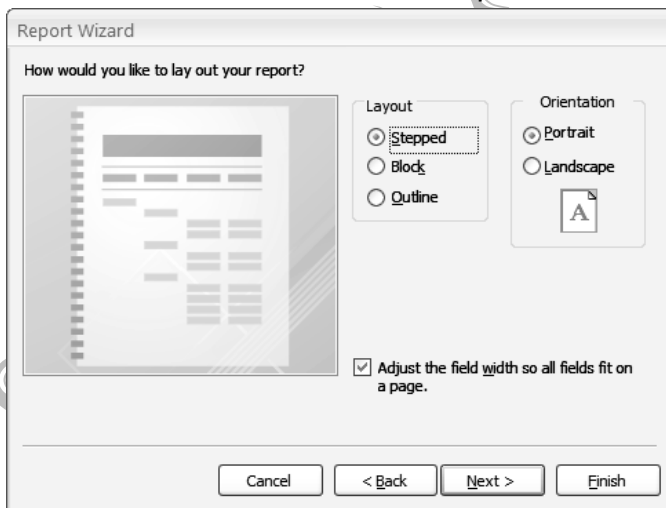
As in Access 2003, the Report Wizard uses the sorting in the table or query that is the source for the report, UNLESS grouping is selected in one of the wizard steps.

If grouping is selected, the report will sort by the field values for the group. In this step, select any additional sorting that is needed.

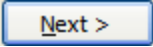
9. Access will sort by the field values for any grouping that was selected. In this step, select any additional sorting that is needed.

In class, sort by **LastName** in Ascending order. Then, sort by **FirstName** in Ascending order.

10. Click on . The next step of the wizard is displayed.

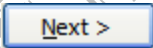


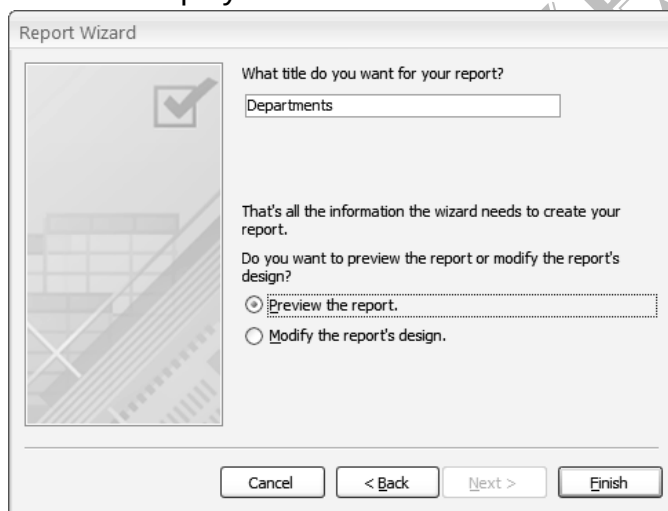
11. Select a Layout, an Orientation and mark the box to place all fields on the page. In class, select a **Stepped** layout a **Portrait** orientation and **mark the checkbox**.

12. Click on . The next step of the wizard is displayed.



13. Click once on a style name in the list box to preview it.

When you find a style for your report, click on . The last step of the wizard is displayed.

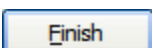


14. Type a title for the report.

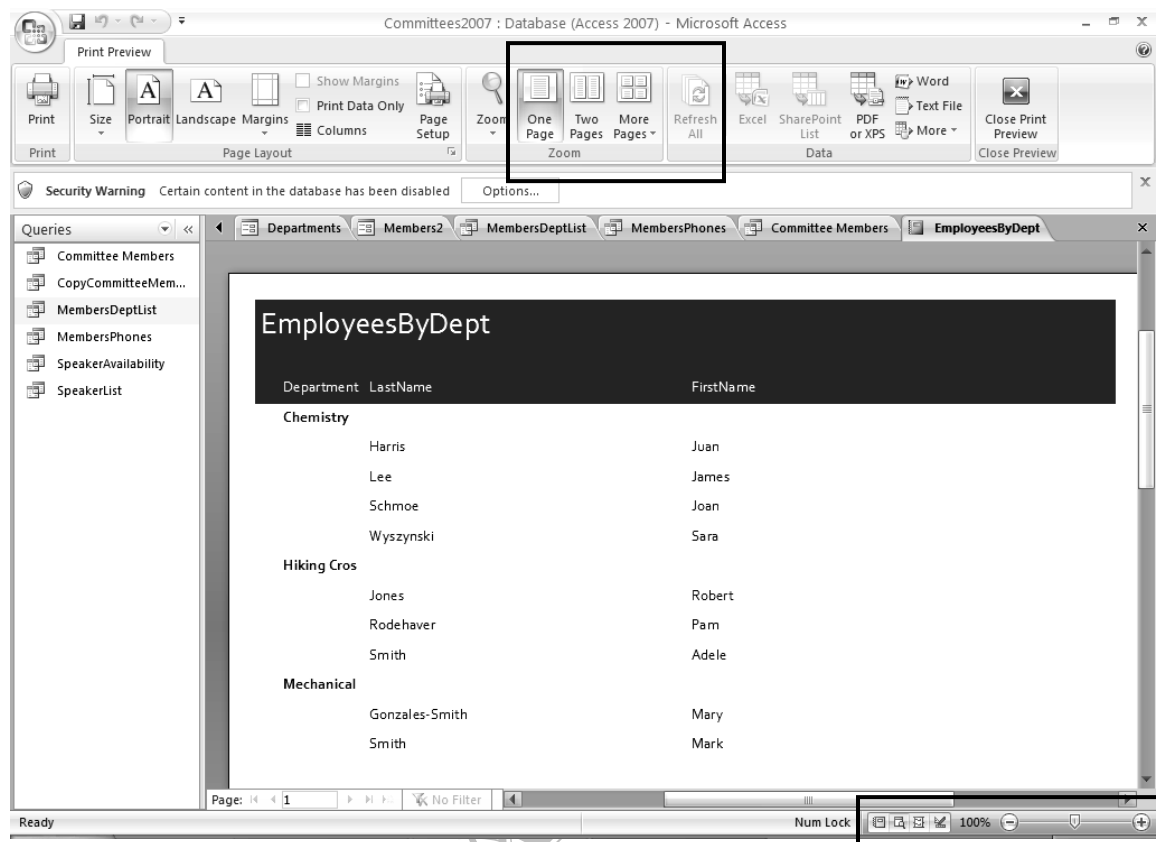
In class, type **EmployeesByDept** as the report title.

15. Mark the option button to Preview the report (Print Preview) or to Modify the report's design (Design view).

In class, mark **Preview the report.**

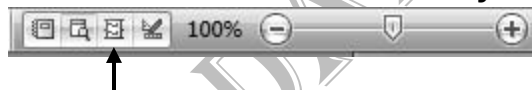
16. Click on .

17. Review the report in Print Preview.



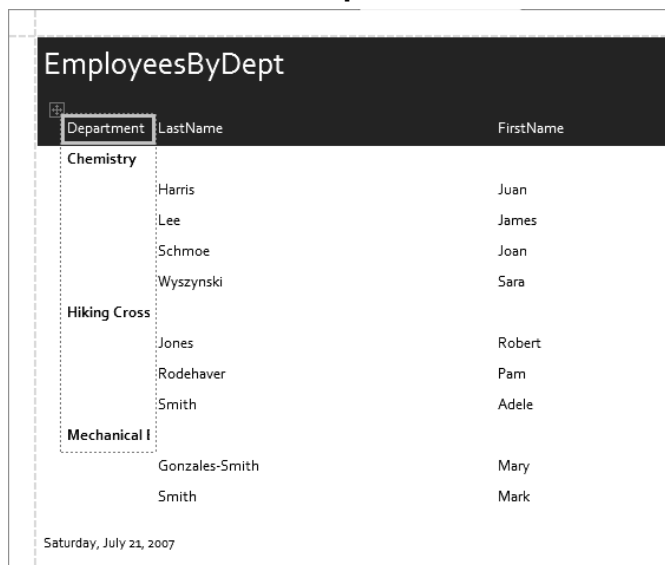
18. Note the Zoom group on the Print Preview tab.

19. Note the View tools on the status bar.

20. On the status bar, click on the **Layout view** tool.

21. Make design changes in Layout view, as you learned in Lesson 4, Edit a Form.

In class, click on the **Department** header to select that column.



Department	LastName	FirstName
Chemistry	Harris	Juan
	Lee	James
	Schmoe	Joan
	Wyszynski	Sara
Hiking Cross	Jones	Robert
	Rodehaver	Pam
	Smith	Adele
Mechanical I	Gonzales-Smith	Mary
	Smith	Mark

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22. To widen a column of the report:

- Place the mouse pointer on the right column border.
 - With the mouse pointer shaped as a two-headed arrow, drag to the right.
- Make the Department column wide enough to display Mechanical Engineering.



Department	LastName	FirstName
Chemistry	Harris	Juan
	Lee	James
	Schmoe	Joan
	Wyszynski	Sara
Hiking Cross-Country	Jones	Robert
	Rodehaver	Pam
	Smith	Adele
Mechanical Engineering	Gonzales-Smith	Mary
	Smith	Mark

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23. Click on the **Save** tool in the Quick Access toolbar to save the changes.

24. In class, do not do this step.

Close the Report by clicking on its X.



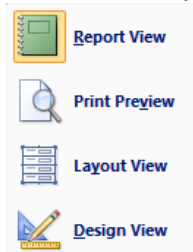
Edit a Report

A report may be edited in one of two views:

- Layout view
- Design view



Use the  tool on the Home tab of the Ribbon to switch between the four main views of a report:

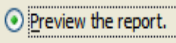


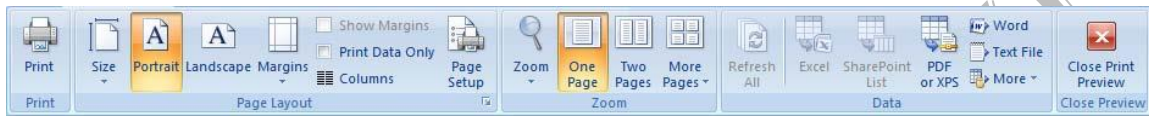
In the previous version of Access, you worked with the Design View and Print Preview for a report. The Layout View allowed you to quickly view some sample records after you made changes to the report's design.

In Access 2007, the Layout View has changed. It allows you to make certain design changes similar to those in the Layout View of a Form. Design View still offers the widest array of changes to a report's design.

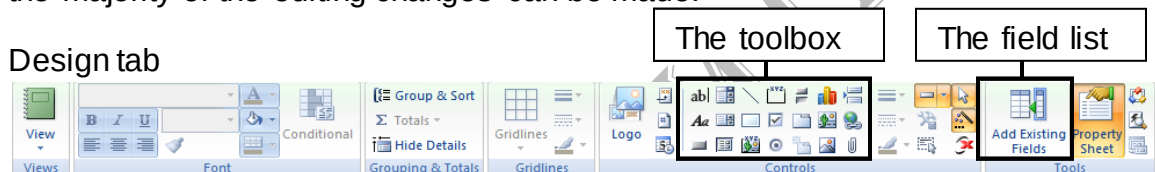
There is a new view, as well - Report View. The report data can be filtered, searched and copied in this view. It is the default view when a report is run from the Navigation Pane.

When a report is created using the Report Wizard, you choose whether to initially view the report in Print Preview or Design view.

- If you select  **Preview the report.**, the Print Preview tab of the Ribbon is displayed. The only changes that may be made in this view are for page layout. The report also may be exported in this view, using the Data group.



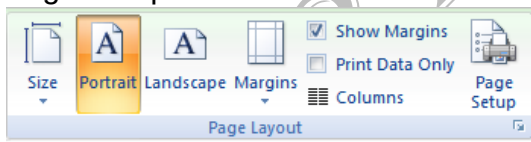
- If you select  **Modify the report's design.**, the three tabs **Design**, **Arrange**, and **Page Setup** of Report Design Tools are displayed. This is the Design view of the report, where the majority of the editing changes can be made.



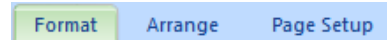
Design tab



Page Setup tab



If Layout view is selected from the View tool, the three tabs of the Report Layout Tools are displayed.

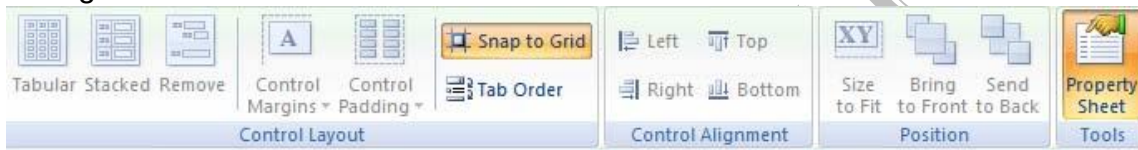


- The Page Setup tab is the same as that available in Design view.
- The Format tab has many of the same tools as the Design tab, but not all.
- The Arrange tab has many of the same tools available in Design view, but the Arrange tab in Design view has more tools.

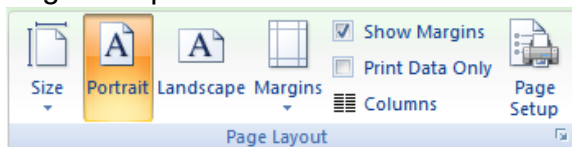
Format tab



Arrange tab



Page Setup tab



E-mail a Report

Discussion

When e-mailing a report, it may be important to select a file format that will preserve the layout and formatting of the original report. These formats are:

- Snapshot format, *.snp (Requires the Microsoft Snapshot Viewer, a free download from the Microsoft Download Center.) At UA, you may install this by opening the Control Panel and Run Advertised Programs.
- Portable Document Format, *.pdf (Requires Adobe Reader)
- XML Paper Specification format, *.xps (Requires the Microsoft SML Paper Specification Viewer.)

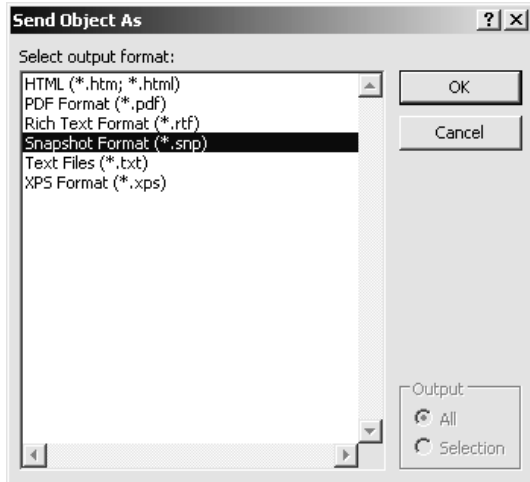
Additional formats are available, but they do not retain the original report's formatting. These formats are:

- HTML, *.htm or *.html (Open in Internet browsers, such as Internet Explorer.)
- Rich Text Format, *.rtf (Open in word processing programs, such as Word.)
- Text, *.txt (Open in word processing and text editing programs, such as Word or Notepad.)

Steps



1. In the Navigation Pane, click once to select the name of the report to be e-mailed.
2. Click on the Office Button.
3. Select E-mail. The Send Object As dialog box is displayed.



4. In the Send Object As box, select the output format to be used.
The discussion on the previous page lists the various formats.
5. Click on OK.
6. Outlook opens a new e-mail message with the report as a file attachment.
7. Complete the e-mail message.
8. Click on Send.

Use Report View



This is a new view for working with reports. In this view, you can filter the data, copy the data and search the data. When a report is run from the Navigation Pane, it defaults to this view.

1. Run the report.

In class, run the report **EmployeesByDept**.

The report is displayed in Report View.

EmployeesByDept

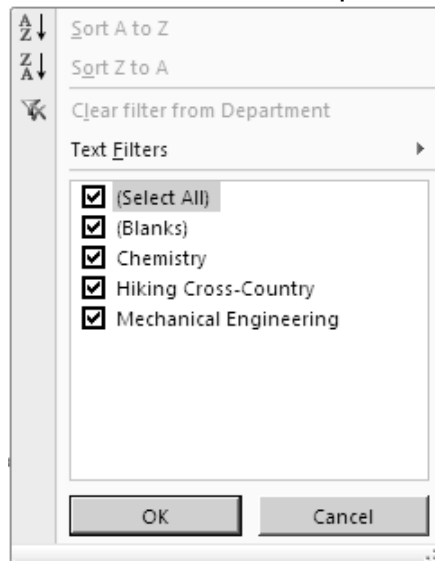
Department	LastName	FirstName
Chemistry		
	Harris	Juan
	Lee	James
	Schmoe	Joan
	Wyszynski	Sara
Hiking Cross-Country		
	Jones	Robert
	Rodehaver	Pam
	Smith	Adele
Mechanical Engineering		
	Gonzales-Smith	Mary
	Smith	Mark

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2. **Click** in one of the field values for Department, such as Chemistry.
The cursor will be displayed.
3. Click on the **Home tab** of the Ribbon to make it the active tab.
4. In the Sort & Filter group, click on the **Filter** tool.



The Filter list for the Department field is displayed.



5. Click on Select All to unmark it.
6. Mark only Chemistry.
7. Click on OK.

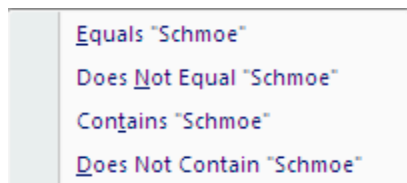
Only the records that meet the criterion are displayed in the report.

EmployeesByDept

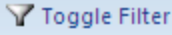
Department	LastName	FirstName
Chemistry	Harris	Juan
	Lee	James
	Schmoe	Joan
	Wyszynski	Sara

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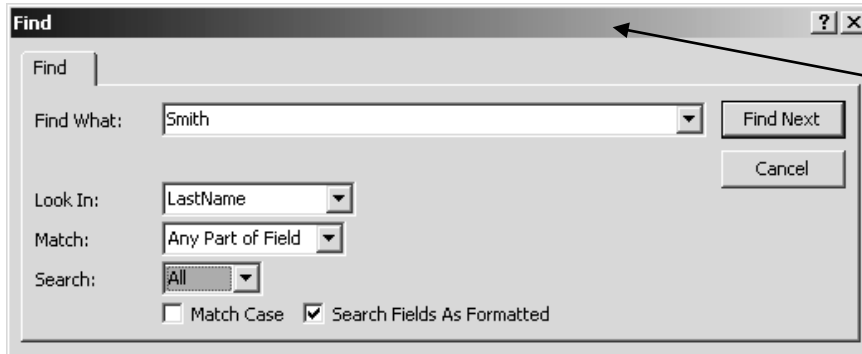
8. Click on Toggle Filter to remove the filter. All the records are displayed.
9. Click in a field value in the LastName field. The cursor is displayed.
10. Click on the Selection tool. A list of options displays.



11. Select Equals "(Name)". The records with that field value in the LastName field are displayed. All other records are hidden.

12. Click on  to remove the filter.
All the records are displayed.

13. With the cursor in the LastName field, click on the Find tool .
The Find dialog box is displayed.
14. Complete the box as shown here.



The Find dialog box is shown with the following settings:

- Find What: Smith
- Look In: LastName
- Match: Any Part of Field
- Search: All
- ☐ Match Case ☒ Search Fields As Formatted

Buttons: Find Next, Cancel

If the dialog box is hiding the data to be viewed, drag the box out of the way by its title bar.

15. Click on Find Next until the message indicates that it has found all the records.
16. Close the Find dialog box by clicking on its X.
17. Drag the mouse across the rows of data for Hiking Cross-Country. The data is selected.

EmployeesByDept

Department	LastName	FirstName
Chemistry		
	Harris	Juan
	Lee	James
	Schmoe	Joan
	Wyszynski	Sara
Hiking Cross-Country		
	Jones	Robert
	Rodehaver	Pam
	Smith	Adele
Mechanical Engineering		
	Gonzales-Smith	Mary
	Smith	Mark

Wednesday, August 08, 2007

18. Click on the Copy tool  on the Home tab.
19. Start Excel.
20. Click in cell A2 to make it the active cell.



21. Click on the  tool on the Home tab of Excel's Ribbon. The data is displayed in the Excel worksheet.

	A	B	C
1			
2	Department	LastName	FirstName
3	Hiking Cross-Country	Jones	Robert
4	Hiking Cross-Country	Rodehaver	Pam
5	Hiking Cross-Country	Smith	Adele
6			

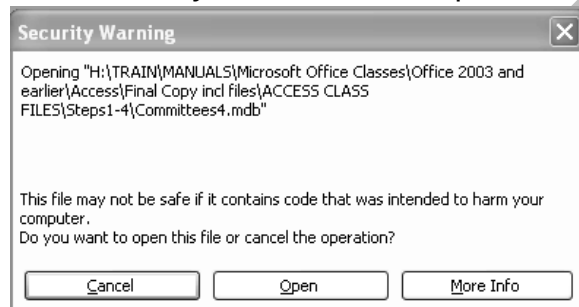
22. Close the workbook and Excel.

Lesson 7: Security

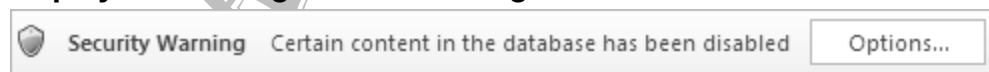
In the previous version of Access, you chose a security level (low, medium or high)



and whether you wanted to run potentially unsafe code or not.



By default in Access 2007, potentially unsafe code is disabled. Then, Access displays a message in the Message Bar.



If you choose to trust the disabled content, there are two choices:

1. Trust the database only for the current session by clicking on the button in the message bar. A dialog box is displayed.

Options...

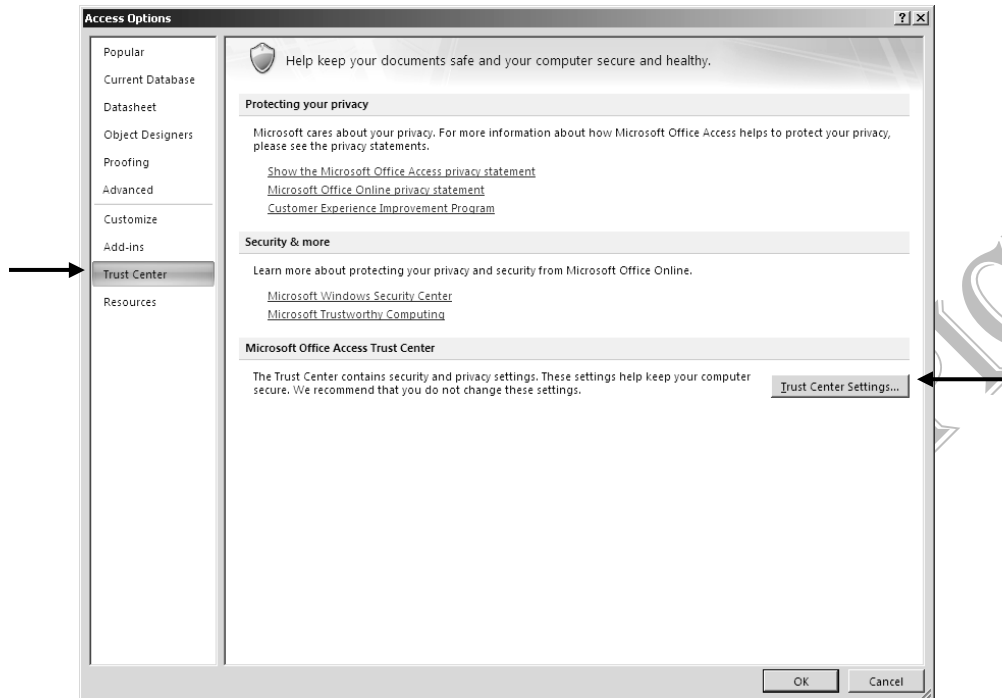
- Select the option Enable this content.



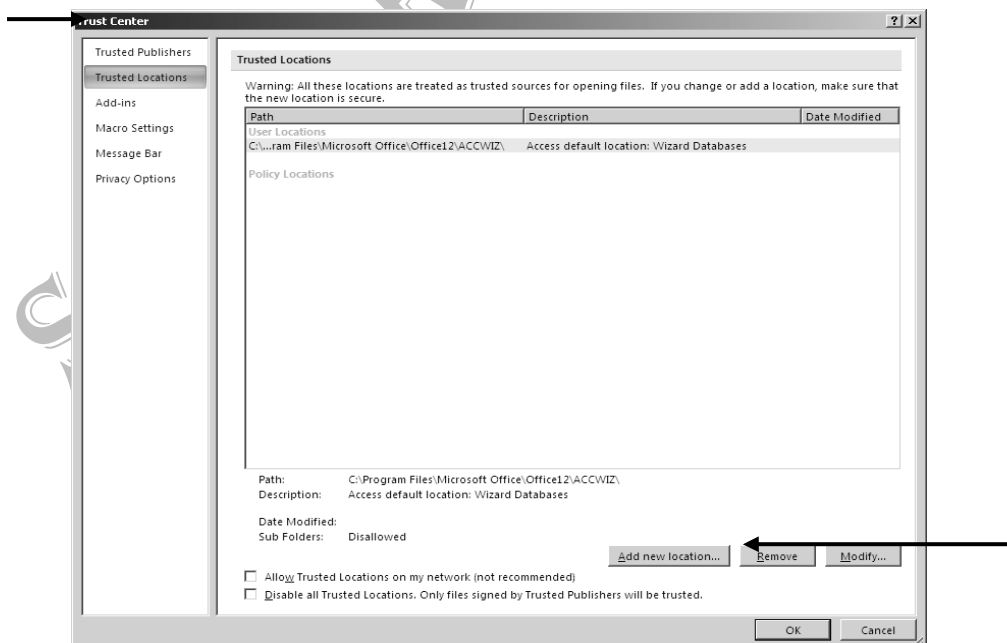
- Click on OK.

OR

2. Trust the database permanently by saving it to a trusted location.
 - To establish a trusted location, click on the Office button and select Access Options.

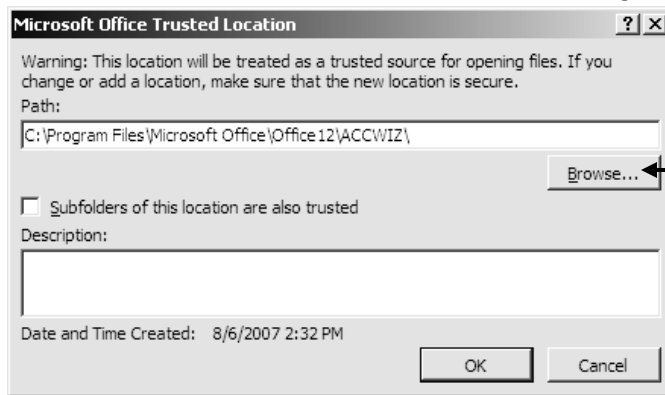


- In the Access Options dialog box, select Trust Center in the left pane.
 - Click **Trust Center Settings...** in the right pane.
- The Trust Center dialog box is displayed.



- In the left pane, select Trusted Locations.
- Click on **Add new location...**.

The Microsoft Office Trusted Location dialog box is displayed.



Tip-Trusted Location
It is not recommended that you trust a location on a network.

- Below the Path edit box, click on Browse to navigate to the drive and folder to be trusted.
- Click OK three times to close all the dialog boxes.
- Move the database to the trusted location or use the Save As command to copy it to the trusted location.

Note: Additional security is provided by requiring a password to open the database. See Appendix B on page 101 for instructions.

Appendix A: Access Options - Suggested Changes to Defaults

To make changes to the defaults:

- Click on the Office Button.
- Click on the Access Options button. The Access Options dialog box is displayed.
- Select **Current Database** from the list in the left column. The options for the open database are displayed in the right column of the dialog box.

Application Options

Application Title:

Application Icon:

☐ Use as Form and Report Icon

Display Form:

☒ Display Status Bar

Document Window Options

☐ Overlapping Windows

☒ Tabbed Documents

☒ Display Document Tabs

☒ Use Access Special Keys ⓘ

☐ Compact on Close

☐ Remove personal information from file properties on save

☒ Use Windows-themed Controls on Forms

☒ Enable Layout View for this database

☒ Enable design changes for tables in Datasheet view (for this database)

☒ Check for truncated number fields

Picture Property Storage Format

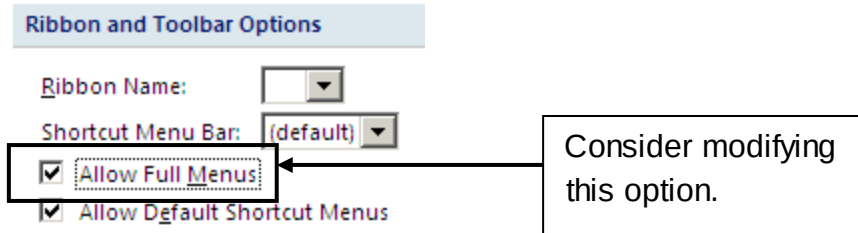
☒ Preserve source image format (smaller file size)

☐ Convert all picture data to bitmaps (compatible with Access 2003 and earlier)

Consider modifying these two options.

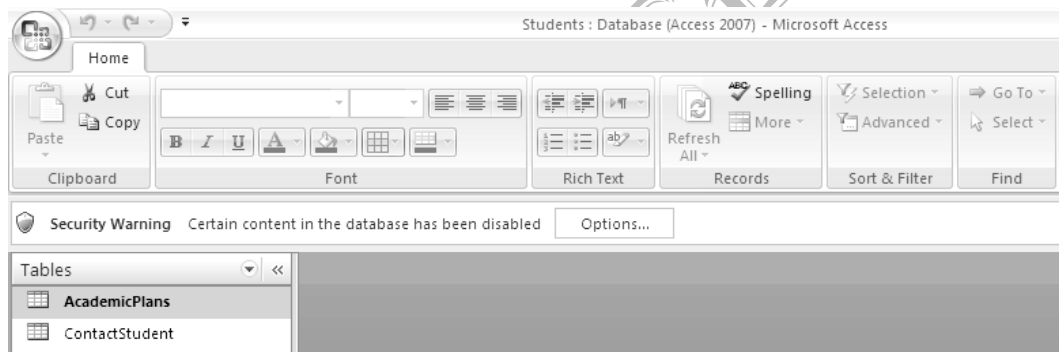
- Removing the mark for **Enable Layout View for this database** prevents accidental changes to objects when they are in the view where data is displayed.
- Removing the mark for **Enable design changes for tables In Datasheet view** prevents accidental changes to tables, which easily are modified in Access 2007. This will remove that Field1 column that is displayed as the last column of a table, for example.

- In the Ribbon and Toolbar Options area of the right column, the following options are displayed.

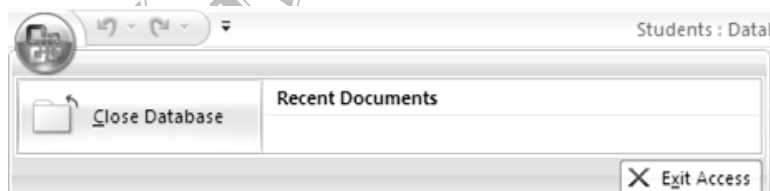


- Removing the mark for **Allow Full Menus** hides all tabs of the Ribbon with the exception of the Home tab. Also, only the Close Database and Exit Access commands are available from the Office Button.
- After the selections are made, click on OK.
- The database must be closed and reopened for the changed options to take effect.

When the database is opened, again, note the changes to the Ribbon.



When the Office Button is selected, only the Close Database and Exit Options are displayed.



IMPORTANT To restore any defaults permanently:

Hold down the shift key when you click on the file name in the Getting Started window of Access to open the database or when you double click on the file name in the Open dialog box.

Click on the Office Button and then Access Options. Make selections and click on OK.
Close and reopen the database.

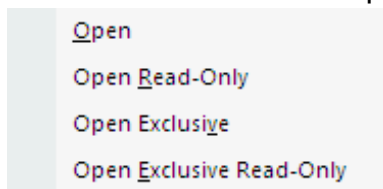
SARDAR AZEEM (PICT)

Appendix B: Use a Password

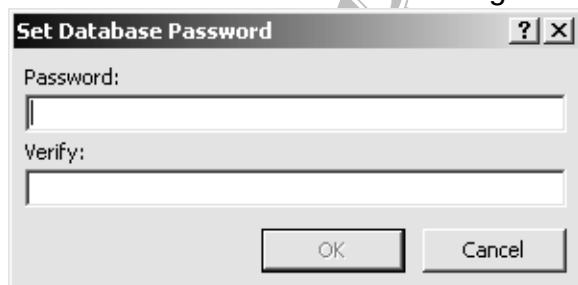
When a password is used to open a database, the database is encrypted when the database is closed and decrypted when it is opened. The database cannot be opened and the data is unreadable unless the password is used.

CAUTION: If the password is forgotten, the database cannot be opened.
The password cannot be retrieved by anyone.

1. Open the database in Exclusive mode.
 - Click the Office Button.
 - Click Open. The Open dialog box is displayed.
 - Navigate to and click once on the filename of the database to open.
 - Select the arrow on the Open button. A list of options is displayed.



- Select Open Exclusive. The database is opened.
2. On the Database Tools tab of the Ribbon, click on  Encrypt with Password. The Set Database Password dialog box is displayed.



3. Type the password in the Password edit box.
4. Type the same password again in the Verify edit box.
5. Click on OK.

When the database is opened in the future, the user will be prompted for a password.

To remove an existing password:

- Open the database using the password.
- On the Database Tools tab of the Ribbon, click on . The Unset Database Password dialog box is displayed.



- Type the password and click on OK.

Appendix C: Open Last Used Database When Access Starts

When the Access 2007 application is opened, Access also can automatically open the last database file that was used.

1. In the Getting Started window or the database window, click on the Office Button.
2. Select the Access Options button. The Access Options dialog box is displayed.
3. In the left column, select Advanced.
4. Scroll down in the right column to the Advanced area.

Advanced

☒ Open last used database when Access starts

Default open mode

☒ Shared

☐ Exclusive

Default record locking

☒ No locks

☐ All records

☐ Edited record

☒ Open databases by using record-level locking

OLE/DDE timeout (sec): 30

Refresh interval (sec): 60

Number of update retries: 2

ODBC refresh interval (sec): 1500

Update retry interval (msec): 250

DDE operations:

☐ Ignore DDE requests

☒ Enable DDE refresh

Command-line arguments:

5. Mark the box Open last used database when Access starts.
6. Click on OK.

Access must be closed and restarted for this option to take effect.

Appendix D: Convert a Database to the Access 2007 File Format

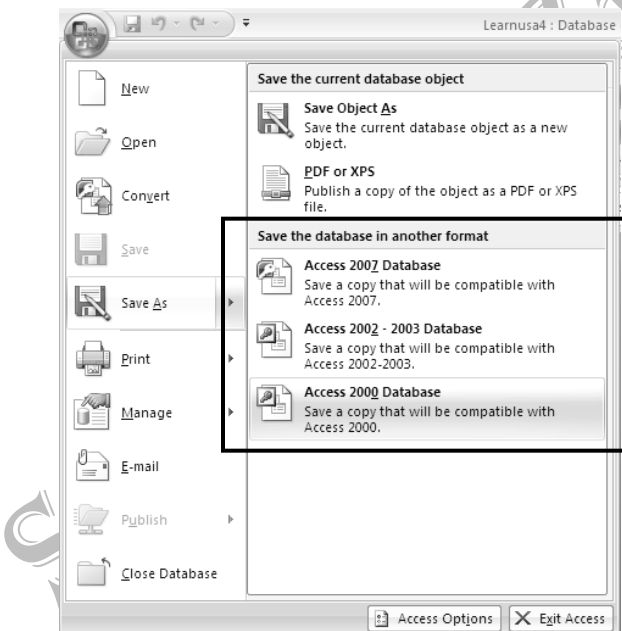
Access 2007 will open, work with and save databases in Access versions 2000, 2002-2003 and 2007. However, a database in the new Access 2007 file format (.accdb) cannot be opened in earlier versions of Access.

If the database is shared with other individuals, ensure that all of them are using Access 2007 before you convert the database to the Access 2007 format.

1. Make a backup copy of the database as a precaution. **DO NOT SKIP THIS STEP.**
 - With the database open, look at the title bar in Access to note the format of the database. For example, this database is in the Access 2000 format.

Learnusa4 : Database (Access 2000 file format) - Microsoft Access

- Click on the Office Button.
- Point to Save As and select the format of the current database, such as Access 2000.



The Save As dialog box is displayed.

- Navigate to the folder and save the database with the word COPY, the same name as the original database file and the current date. (Remember that the slash characters are not allowed. Use hyphens for the date.)

For example, the backup copy of the **LearnUSA database** might be

Learn USa Copy

2. Click on the Office Button.
3. Point to Save As and select the Access 2007 file format. The Save As dialog box is displayed.
4. Navigate to the appropriate folder and type a file name for the database or accept the same name.

If the database name is the same, such as LearnUSA, there will be a database named LearnUSA.mdb (mdb is the extension for database in previous versions of Access) and LearnUSA.accdb (accdb is the extension for databases in the Access 2007 file format).

5. When you click on Save, Access will:
 - Close the original database
 - Create a copy of the database in the new file format
 - Open the database in the new file format

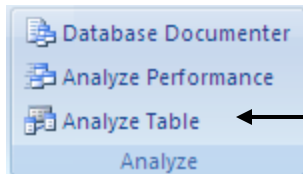
When a database in the Access 97 format is opened, the Database Enhancement dialog box appears. Click on Yes to begin the conversion process.

To convert a database in the Access 95 format, open the database in Access 2000, 2002 or Access 2003 and save it in one of those formats. Then, open the database in Access 2007 and follow the instructions above

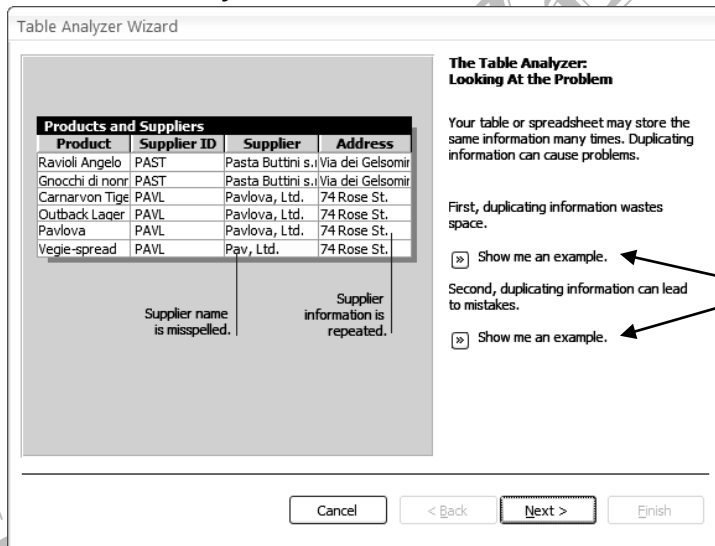
Appendix E: Table Analyzer Wizard

If you are constructing a database by importing data from other sources, the Table Analyzer Wizard in Access can help to identify repeating data. The wizard provides a method to organize the repeating data into separate tables, so that the data is stored efficiently. Access preserves the original tables of imported data as a backup copy and creates new tables for your database. This process is called normalization.

1. Open the database, if it is not open.
2. Close the table with the imported data if it is open.
3. It is a best practice to click on the Office Button and use the Save As command to save a copy of the database, before you begin to modify it.
4. On the Database tools tab in the Analyze group, click on Analyze Table.



The Table Analyzer wizard starts.



For more information, click on the  tool labeled **Show me an example.**

5. Work your way through the steps of the wizard. In addition to the  tool, there are additional tools to assist you that are available within the wizard.



Tips



Rename



Undo



Set Unique Identifier (Primary key)



Add Generated Key (Access creates an ID field that is auto-numbered and is primarykey.)

Appendix F: DATABASE CONCEPTS

Data

Data are any facts, numbers, or text that can be processed by a computer. Today, organizations are accumulating vast and growing amounts of data in different formats and different databases.

In computer science, **data** is anything in a form suitable for use with a computer. Data is often distinguished from programs. A program is a set of instructions that detail a task for the computer to perform. In this sense, data is thus everything that is not program code.

This includes:

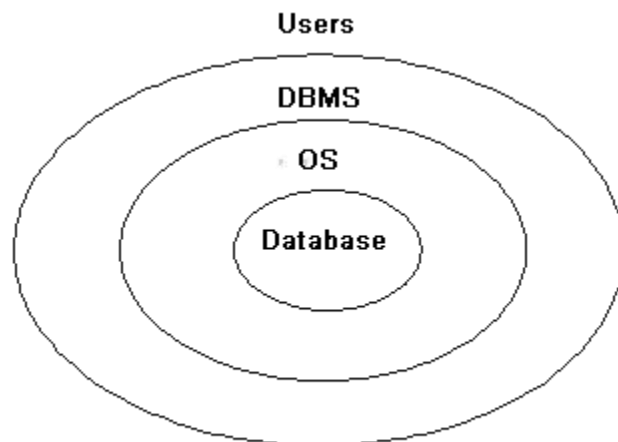
- operational or transactional data such as, sales, cost, inventory, payroll, and accounting
- nonoperational data, such as industry sales, forecast data, and macro economic data
- meta data - data about the data itself, such as logical database design or data dictionary definitions

Database

A **database** is a collection of information that is organized so that it can easily be accessed, managed, and updated. In one view, databases can be classified according to types of content: bibliographic, full-text, numeric, and images.

In computing, databases are sometimes classified according to their organizational approaches. The most prevalent approach is the relational database, a tabular database in which data is defined so that it can be reorganized and accessed in a number of different ways. A distributed database is one that can be dispersed or replicated among different points in a network. An object-oriented programming database is one that is congruent with the data defined in object classes and subclasses.

As one of the oldest components associated with computers, the **database management system (DBMS)**, is a computer software program that is designed as the means of managing all databases that are currently installed on a system hard



drive or network. Different types of database management systems exist, with some of them designed for the oversight and proper control of databases that are configured for specific purposes. Here are some examples of the various incarnations of DBMS technology that are currently in use, and some of the basic elements that are parts of DBMS software applications.

In database management system (DBMS), data files are the files that store the database information, whereas other files, such as index files and data dictionaries, store administrative information, known as metadata.

Database system is a system to achieve an organized, store a large number of dynamical associated data, facilitate for multi-user accessing to computer hardware, software and data, that it is a computer system with database technology.

Why use electronic databases?

Electronic Databases are the major sources of information. Increasingly, information is being published only in electronic formats. There is coverage in virtually all areas of knowledge: science, engineering, mathematics, medicine, agriculture, psychology, sociology, philosophy, law, business, economics, education and more. A single database may refer to a variety of sources, including periodical articles, books, government documents, industry reports, papers at meetings, newspaper items, films, video recordings etc.

With a terminal with Internet connection you can be seconds away from receiving valuable information on any imaginable topic. These rich and voluminous databases are stored in by the information companies or agencies known as retrieval services. Once you are connected to them, much of the world's knowledge literally will be at your fingertips. As an information source a database may be shared by thousands of users simultaneously, and it is available whenever the retrieval service is in operation. There is no limit to the number of times a database can be searched or the number of times an item can be displayed. Unlike a library book, databases do not deteriorate physically, nor can they be misplaced, stolen, or vandalized.

Table Definition

A database object that stores information that relates to one entity. For example, if the entity was "Customer", the data stored in the "Customer" table might include the following:

First Name

Last Name

Office Phone

Contact Name

If the entity was "Product", the data stored might include the following:

Product ID

Product Name

Full Description

Pre pared by: Sardar Azeem(MBA(B&F) Computer HW And Network Engineer: Pict Computer

Center Link Road Abbottabad.

P a g e | 109

Azeem_aag2000@yahoo.com

Price

Supplier

Relationship

A relationship, in the context of databases, is a situation that exists between two relational database tables when one table has a foreign key that references the primary key of the other table. Relationships allow relational databases to split and store data in different tables, while linking disparate data items.

Relationship types

You share many relationships with members of your family. For instance, you and your mother are related. You have only one mother, but she may have several children. You and your siblings are related—you may have many brothers and sisters and, of course, they'll have many brothers and sisters as well. If you're married, both you and your spouse have a spouse—each other—but only one at a time. Database relationships are very similar in that they're associations between tables. There are three types of relationships:

- **One-to-one:** Both tables can have only one record on either side of the relationship. Each primary key value relates to only one (or no) record in the related table. They're like spouses—you may or may not be married, but if you are, both you and your spouse have only one spouse. Most one-to-one relationships are forced by business rules and don't flow naturally from the data. In the absence of such a rule, you can usually combine both tables into one table without breaking any normalization rules.
- **One-to-many:** The primary key table contains only one record that relates to none, one, or many records in the related table. This relationship is similar to the one between you and a parent. You have only one mother, but your mother may have several children.
- **Many-to-many:** Each record in both tables can relate to any number of records (or no records) in the other table. For instance, if you have several siblings, so do your siblings (have many siblings). Many-to-many relationships require a third table, known as an associate or linking table, because relational systems can't directly accommodate the relationship.

Appendix G: DATABASE FEATURES

Attribute

Definition: A single data item related to a database object. The database schema associates one or more attributes with each database entity.

Also Known As: field, column

Records

Definition: A database record consists of one set of tuples for a given relational table. In a relational database, records correspond to rows in each table.

Tuple

Definition: In the context of databases, a tuple is one record (one row).

Keys

Databases use tables to organize information. (If you don't have a basic familiarity with database concepts, read What is a Database?) Each table consists of a number of rows, each of which corresponds to a single database record. So, how do databases keep all of these records straight? It's through the use of keys.

Primary Keys

A Key that uniquely identifies a record in a database. Primary key has two main features first it is unique and other it is always not null. Every database table should have one or more columns designated as the primary key. The value this key holds should be unique for each record in the database. For example, assume we have a table called Employees that contains personnel information for every employee in our firm. We'd need to select an appropriate primary key that would uniquely identify each employee. Your first thought might be to use the employee's name.

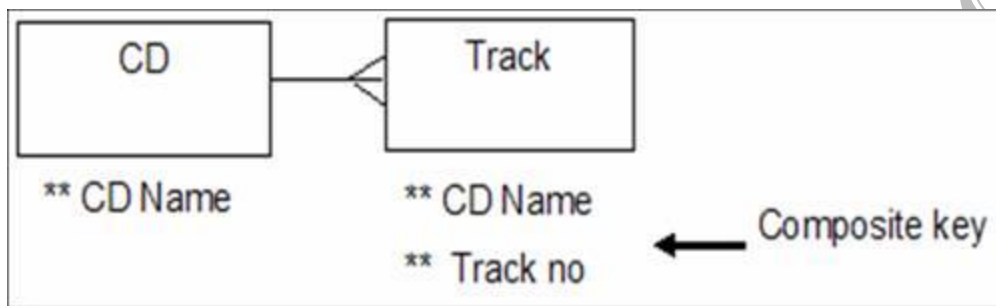
Foreign Keys

A key which is primary key in one table and non-key in the other table. These keys are used to create relationships between tables. Natural relationships exist between tables in most database structures. Returning to our employees database, let's imagine that we wanted to add a table containing departmental information to the database. This new table might be called Departments and would contain a large amount of information about the department as a whole. We'd also want to include information about the employees in the department, but it would be redundant to have the same information in two tables (Employees and Departments). Instead, we can create a relationship between the two tables.

Composite Keys

A composite key consists of more than one attribute to uniquely identify an entity occurrence. This differs from a compound key in that one or more of the attributes, which make up the key, are not simple keys in their own right.

For example, you have a database holding your CD collection. One of the entities is called tracks, which holds details of the tracks on a CD. This has a composite key of CD name, track number.



Referential Integrity

Definition: Referential integrity is a database concept that ensures that relationships between tables remain consistent. When one table has a foreign key to another table, the concept of referential integrity states that you may not add a record to the table that contains the foreign key unless there is a corresponding record in the linked table. It also includes the techniques known as cascading update and cascading delete, which ensure that changes made to the linked table are reflected in the primary table.

Consider the situation where we have two tables: Employees and Managers. The Employees table has a foreign key attribute entitled Managed By which points to the record for that employee's manager in the Managers table. Referential integrity enforces the following three rules:

1. We may not add a record to the Employees table unless the Managed By attribute points to a valid record in the Managers table.
2. If the primary key for a record in the Managers table changes, all corresponding records in the Employees table must be modified using a cascading update.
3. If a record in the Managers table is deleted, all corresponding records in the Employees table must be deleted using a cascading delete.

Use Of wild Cards

A wildcard is a special character that can represent one or more characters in a text value. You can use wildcards to find many records with similar, but not exactly the same, information. You can also use them to look for a specific record when you can't remember enough information to retrieve just that one record.

The ability to find and retrieve data easily is great, but use wildcards cautiously. If you use the wrong character, a query will almost certainly return erroneous data. If you're not lucky enough to catch the mistake visually, the erroneous data could go undetected for a long time--long enough to corrupt your data or even your entire application.

1: Match characters in a specific position

The most flexible wildcard character is the asterisk (*). It matches any character or any block of characters in a specific position. For instance, the following statement would return any entry that contains the string *access* without regard to letter case:

Like `"*Access"`

So it would return the following entries: *Microsoft Access*, *Access 97*, and *accessing*.

Drop the first * character to match entries that begin with the string *access*:

Like `"Access"`

This statement would return *Access 97* and *accessing* but not *Microsoft Access*. Similarly, the following statement would return *Microsoft Access* but not *Access 97* or *accessing*:

Like `"*Access"`

2: Spaces matter in a * match

When using the * character to match characters in a specific position, pay close attention to space characters. If the search string includes a space character between the literal characters and the * character, Access will return only those entries that include a space character in that position. For instance, the following statement matches *Microsoft Access* but not *Access 97* or *accessing*:

Like `"* Access"`

You might not expect that search string to return *Access 97*, but the absence of the word *accessing* might be a surprise. Access won't return *accessing* because the entry doesn't have a space character before the *access* block. Similarly, the following statement matches *Access 97* but not *Microsoft Access* or the single word entry, *accessing*:

Like `"Access *"`

3: Match just one character in a specific position

The question mark character (?) serves as a single-character placeholder. You can combine this wildcard to specify a certain number of characters. For instance, the following statement would match *Smith* and *Smyth* but not the word *smooth* (without regard to letter case):

Like "Sm?th"

You can combine ? characters to cast a wider net or to focus the search, depending on the circumstances. For example, the following criteria would find *Smith*, *smile*, *smite*, and *smirk*:

Like "Smi??"

It would not find *smirch* or *smitten*. That's because each ? character represents only one character. Although both *smirch* and *smitten* match the literal characters in the expression (*smi*), both words have more than two characters following those literal characters. To catch all entries that begin with *smi* without regard to the number of characters, use the * character (# 1).

4: Use ? to match symbol characters

Although the English language doesn't use special characters, like è and ü, it's common to store values that contain such characters. They pose a unique problem in a search, because you must use the special character and not just the English counterpart. You might find it easier to use the ? placeholder in these cases. For instance, if you want to find the record for *Comércio Mineiro*, you could take the time to insert the Ã© character, but doing so requires extra keystrokes. Instead, use the ? wildcard to represent the special character, as follows:

Like "Com?rcio *"

This search expression might match a few other entries, but in most cases, it won't, because it's such a narrow search. In this example, you must also include the * or the literal string to match the desired entry because it contains more than one word. If you omitted the *, Access would match an entry that contained only the text *Com?rcio*.

5: Use [] to match literal wildcard characters

Sometimes, you'll want to include a wildcard character as part of a literal search string. For instance, you might use the * character to denote a special note or comment in a memo or text field. In that case, you must tell Access to evaluate the * character as a literal character rather than a wildcard by enclosing the wildcard in brackets as follows

Like "[*]*"

If you use *Like* "*", Access will return every entry in the data source. (It isn't necessary to enclose the ! or] wildcard characters in brackets.)

6: Match characters in a list

It's possible to match more than one specific character at a time using a list. To match any single character in a list, separate each item in the list with a comma and enclose the list of characters in brackets. For instance, the following search string will find any entry that begins with a or z:

Like "[a, z]*"

If the list is a set of consecutive values, use the hyphen to separate the first and last characters in the sequence. The following expression matches all entries that begin with the letters a, b, c, d, or e:

Like "[a-e]*"

The bracket wildcards are flexible enough to handle multiple lists as follows:

Like "[a-e, k, p-s]*"

The above expression matches all entries that begin with the following letters: a, b, c, d, e, k, p, q, r, or s.

You can also use the bracket wildcards to denote characters in the middle of a string. For instance, the following expression matches entries that begin with the letter a and have the letters b or f as the second character:

Like "a[b, f]*"

7: Exclude characters

Most of the wildcard characters match characters. The exclamation point character (!) is different. Use ! to exclude matching characters from a query's results. For instance, the following expression would return all entries that do *not* begin with the letter a:

Like "[!a]*"

Notice that the expression includes the bracket wildcards. The ! character works with the brackets. In truth, you can exclude matches only from a list. However, the list can contain just a single character. Expand the list possibilities by applying the same techniques discussed in # 6.

Prepared by:

**SARDAR AZEEM(MBA(B&F)
COMPUTER HW AND NETWORK
ENGINEER**

**PICT COMPUTER CENTER LINK ROAD
ABBOTTABAD.**

Define:

Success;
it's when your
signature becomes
an
autograph